A Hybrid Decomposition-Based Deep Learning Approach for Enhanced Wind Power Forecasting

by

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Abstract

The massive demand for fossil fuels has negatively impacted our climate. Renewable energy sources, such as wind, solar, and hydraulic energy, have recently become an excellent choice for sustainable and climate-friendly energy. Wind turbine energy generators have become an excellent solution in this area. However, the non-stationary nature of the wind presents challenges in achieving high levels of efficiency. To address this issue, this study has proposed a method that considers a hybrid prediction model integrating three pre-processing and prediction modules, which are a Complete Ensemble Empirical Mode Decomposition with an Adaptive Noise (CEEMDAN) module, an Empirical Wavelet Transform (EWT) module, and a Gated Recurrent Unit (GRU) module. Those components have been utilized to process the data by reducing the noise and providing stability to improve the GRU model's captivity in the prediction. Furthermore, the proposed integrated CEEMDAN-EWT-GRU model has been evaluated in three structured experiments, each with unique objectives to see the effectiveness of the proposed method. The first experiment consists of evaluating the proposed method against baselines of deep learning models, benchmark [17], and a hybrid model that integrates CEEMDAN and EWT across time horizons before selecting the optimal time horizons that will be selected for further investigation. The second experiment evaluates the performance of the proposed method by integrating more features, which is a novelty in this work. In the third experiment, we introduce a hybrid method, CEEMDAN-EWT-GRU-BiLSTM, which applies CEEMDAN and EWT to the original data. The Intrinsic Mode Functions (IMFs) will be divided into high-frequency and low-frequency IMFs using the permutation entropy. The GRU will be trained on high-frequency IMFs, while Bi-LSTM will be trained on low-frequency IMFs. The findings of this research confirm that the integration of the decomposition technique has provided superior accuracy and robustness compared to the standalone deep learning prediction models. The GRU has shown more suitability for integration with the CEEMDAN and EWT for real-time forecasting applications, especially in wind energy systems. This work has a number of contributions to wind power forecasting: (1) introducing a more efficient method CEEMDAN-EWT-GRU for short-term forecasting, (2) demonstrating the effectiveness of the GRU against LSTM and Bi-LSTM, (3) evaluating the effect of integrating more features into the proposed method, (4) introducing the hybrid method of GRU and Bi-LSTM based on the frequency of IMFs, and (5) evaluating the performance of the methods for varying historical and forecasting horizons.

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I am grateful for his feedback and guidance, which helped contribute to AI and influenced me towards pushing my limit in this field.

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Dedication

To my beloved parents,

I dedicate this work to you with warm appreciation. Your support, love, and encouragement were my most significant sources of strength during my master's program. Your belief in my abilities gave me the necessary resources and opportunities to excel in my career.

Thank you for being by my side. This achievement is as much yours as it is mine.

With all my love and appreciation, this is for you.

Table of Contents

List of Tables

List of Figures

List of Abbreviations

AI	Artificial Intelligence.
ML	Machine Learning.
CEEMDAN	Complete Ensemble Empirical Mode Decomposition with Adaptive Noise.
EMD	Empirical Mode Decomposition.
EEMD	Ensemble Empirical Mode Decomposition.
WD	Wavelet Decomposition.
EWT	Empirical Wavelet Transform.
GRU	Gated Recurrent Unit.
RNN	Recurrent Neural Networks.
LSTM	Long Short-Term Memory.
Bi-LSTM	Bi-Directional LSTM.
ML	Machine learning.
DL	Deep learning.
GP	Gaussian Process.
SVM	Support Vector Machines.
RF	Random Forests.
KNN	K-Nearest Neigbors.
NN	Neural Network.
RBFNN	Radial Basis Function Neural Network.
ARMA	Autoregressive Moving Average.
NWP	Numerical Weather Prediction.
ARD	Automatic Relevance Determination.
RBE	Rank Bayesian Ensemble.
KDE	Kernel Density Estimation.
sVAR	Sparse Vector Autoregressive.
ARFIMA	Autoregressive Fractional Integrated Moving Average.
IMFs	Intrinsic Mode Functions.
RMSE	Root Mean Square Error.
MAE	Mean Absolute Error.

Chapter 1

Introduction

1.1 Preface

Humanity has reached a high level of advancement in technology and infrastructure with the rapid population growth. This advancement is linked with resources such as food, shelter, transportation, and electricity, which are essential for maintaining modern living standards. However, we have long relied heavily on nonrenewable energy. These energies have raised two main concerns. First, this energy accelerates climate change due to the significant carbon dioxide emissions. Second, there is an increase in the depletion of resources due to the limited amount of nonrenewable resources. This has led the government to shift towards renewable energy due to clean energy and natural availability. They increasingly turn to renewable energy options such as wind, solar, hydropower, and geothermal. These renewable energy sources harness natural elements through human innovation, such as solar panels and wind turbines, to generate clean energy with a significantly lower environmental impact than fossil fuels.

Wind energy provides benefits such as producing clean energy. However, wind energy has a key challenge is high variability, which makes prediction difficult. The researchers focus on improving prediction accuracy through enhanced methods.

1.2 Motivation

Integrating renewable energy into power systems must be achieved using an accurate wind power forecasting method. The wind power data contains high variability and nonstationary, and the conventional methods fail to deal with those situations; this demands advance methods to handle the challenges. The section highlighted the key motivations of this paper:

- **Importance of Accurate Forecasting for Renewable Energy Integration.** Accurate wind power forecasting helps maintain main grid stability, optimize energy use, and support participation in the energy market. This will make wind energy important to integrate into energy systems. Providing an accurate forecasting method to overcome the challenges of wind power's unpredictable nature will provide a huge boost for the use of renewable energy systems.

- **Challenges of Nonstationary and Nonlinear Data.** The wind data contains complex patterns such as nonstationary, non-linear, and noise. Both the statistical methods [2] and traditional machine learning [3] fail to handle those challenges, which leads to poor performance.

- **Limitations of Existing Deep Learning Models.** RNNs [36] and LSTMs [31] are types of deep learning models being used for wind power forecasting. Those models have issues such as vanishing gradients and poor performance when handling noisy data, this lead to build a hybrid method to deal with the challenges of wind power forecasting.

- **Potential of Signal Decomposition for Enhanced Feature Extraction.** The advanced signal decomposition techniques, such as CEEMDAN (Complete Ensemble Empirical Mode Decomposition with Adaptive Noise) [28] and Empirical Wavelet Transform (EWT) [21], offer powerful tools for reducing noise, and extracting meaningful features. Integrating these techniques with the machine learning method would significantly improve the prediction accuracy.

- **Need for Optimized Forecasting Configurations.** The effectiveness of forecasting models often depends on the choice of historical and forecasting lengths. The optimal configurations were selected to ensure that the model achieved accurate prediction.

- **Practical Implications for Renewable Energy Management.** Demand an effective forecasting method to maintain grid stability and optimize scheduling and market operations. The accurate forecasting method contributes to less demand for nonrenewable energy, reduces energy waste, and supports the integration of clean energy.

1.3 Problems and Challenges

Wind power data contains nonstationary, nonlinear, and noise, which causes challenges for methods to handle those challenges effectively. Those current methods are poorly effective in dealing with the complex of the wind power data, leading to poor performance. This requires a hybrid method to deal with nonstationary and noise. Hence, research studies utilize both WD and EMD [23], but the result is not satisfied since these techniques have limitations such as mode-mixing in EMD. These challenges have demanded the utilization of an advanced method with deep learning to enhance wind power forecasting.

1.4 Proposed Solution

The purpose of this research paper is to develop a hybrid method that effectively addresses the challenges and improves prediction accuracy. This paper introduces a hybrid framework utilizing CEEMDAN, EWT, and GRU models. The CEEMDAN decomposes the original signal, EWT reduces the noise of the high-frequency IMF, and the GRU is used to predict the wind power by processing the IMFs. This proposed framework has effectively handled the challenges and enhanced the performance of wind power forecasting.

1.5 Contribution

The key contributions of this thesis are summarized in the following.

- **Development of a Hybrid Forecasting Framework**: The paper proposed a hybrid framework utilizing CEEMDAN, EWT, and GRU. The framework is built to enhance the input quality by extracting meaningful features, reducing noise, and enhancing the prediction accuracy.

- **Comprehensive Evaluation of Input and Output Lengths**: The paper has done extensive experiments conducted with different input and output lengths to identify the optimal configurations of wind power forecasting methods. These experiments provide insights into how different configurations affect the prediction accuracy.

- **Benchmarking Against State-of-the-Art Models**: This work rigorously compares the proposed methodology with state-of-the-art models, including RNN, LSTM, GRU, Bi-LSTM, benchmark and hybrid models. The methods have been evaluated using multiple quantitative metrics, such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-squared (R2), and Margin of Error (ME).

- **Investigating the Impact of Additional Meteorological Features**: The paper has explored adding features into the hybrid framework to observe the prediction performance.

- **Hybrid GRU-BiLSTM Model Based on Frequency Decomposition**: The paper has introduced a GRU-BiLSTM model using a split of IMFs to enhance the prediction accuracy compared to the proposed method.

- **Contribution to Renewable Energy Research**: The paper's purpose is to provide an enhanced forecasting framework and contribute to advancing renewable energy research.

1.6 Organization of Thesis

The structure of this thesis is outlined as follows: Chapter 2 provides a comprehensive review of current methodologies for wind power forecasting. Chapter 3 gives a comprehensive overview of the proposed hybrid forecasting framework. Details the integration of CEEMDAN and EWT together with a GRU model. The chapter discusses the decomposition process, feature extraction, and the architecture of the GRU model. Chapter 4 focuses on the design, implementation and analysis of experiments to evaluate the proposed framework and discussions. Finally, Chapter 5 summarizes the findings of this thesis.

Chapter 2

Background

This chapter provides an overview of the wind power forecasting method. Section 2.1 consists of two methods of wind power: physical models and statistical methods. Section 2.2 is subdivided into two areas on the field of machine learning. Section 2.3 explores decomposition techniques for wind power forecasting.

2.1 Conventional Wind Power Forecasting Approaches

Before the introduction of ML in wind power forecasting, several methods, such as physical models and statistical methods, were used, each with advantages and disadvantages.

2.1.1 Physical Models

NWP models are physical models that leverage atmospheric and physical laws for forecasting. Liu et al. [20] combined these techniques to enhance ultra-short-term wind power forecasting by using the Rank Bayesian Ensemble (RBE) and a fluctuation awareness model. These methods stabilize NWP output by incorporating probabilistic fluctuation scenarios. In addition, Chen et al. [5] utilized Gaussian processes (GP) to correct NWP predictions, integrating meteorological variables such as wind speed, direction, temperature and pressure and employing Automatic Relevance Determination (ARD) for feature selection. Another paper by Hu et al. [13] developed a hybrid approach that combines corrected NWP data with spatial correlations between neighboring wind farms, optimizing GP models with combinations of kernel functions to improve prediction accuracy. Khalid et al. [18] proposed a work that enhances short-term wind power prediction by utilizing the NWP data and multiple observation points within a wind farm. Their method is used to predict wind speed and power using autoregressive modeling at 10 resolution. However, these studies have limitations. The NWP requires high dimensional data and does not effectively integrate the historical data; these highlighted the need for hybrid methods using advanced methods.

2.1.2 Statistical Methods

Statistical methods for wind power prediction commonly employ autoregressive models and hybrid approaches to manage the temporal dependencies found in wind speed data effectively. Torres et al.[27] utilized Autoregressive Moving Average (ARMA) to predict the hourly wind speed in Spain by combining data transformation and standardization to handle the nonstationary and seasonality. Their method was promising by obtaining accurate predictions for long-term forecasts. However, the performance of the ARMA model dropped for predictions extending past two hours because of its fundamental linear assumptions. Hill et al. [12] utilized autoregressive (AR) and multivariate autoregressive models to examine daily and seasonal fluctuations in wind speed data from the UK. The method has effectively captured the spatial correlation across various regions. However, the study have issue with handling the complexity of their method for large geographic areas. Han et al. [11] built a hybrid method that combine both ARMA and Kernel Density Estimation (KDE), this method can distinguish between linear and nonlinear elements which it enhances the prediction accuracy. However, this method has introduced computational costs. Dowell et al. [8] has utilize Sparse Vector Autoregressive (SVAR) to predict the wind power forecasting for short-term. This model has taken advantage of the spatio-temporal connections between various wind farms. Sadaei et al. [24] has introduce a hybrid method combine the autoregressive fractional integrated moving average (ARFIMA) and fuzzy time series (FTS). This method can capture both the short and long-term memory. To enhance the prediction accuracy, this required parameter tuning using particle swarm optimization. However, this step introduced complexity in finding the best parameters. Additionally, those techniques have drawbacks such as scalability and computational efficiency and are not well suited for non-linear relationships.

2.2 Machine Learning for Wind Power Forecasting

Ml has significantly replaced the conventional method such as physical modeling and statistical, by directly learn from data. This allows the ML method to capture complex patterns such as learn nonlinear relationship. These techniques have been extensively used in various domains, such as wind power forecasting, because of their capacity to handle large datasets and reveal hidden patterns. Machine learning is classified into two categories: traditional machine learning, which is based on statistical and algorithmic methods, and deep learning models, which employ multi-layered neural networks to learn layered representations of data. Each of these methodologies presents distinct advantages, making them essential tools for tackling the challenges associated with wind power prediction.

2.2.1 Traditional Machine Learning Models

Traditional machine learning models have been introduced for wind power prediction because of their ability to capture the nonlinear relationship data, which the statistical method fails to do. Jorgensen et al. [16] have done an extensive analysis of the state-of-the-art of models such as SVM, KNN, RF and NN for wind power prediction. The paper has stated that SVM and NN are the most common because NN excels in capturing non-linear relationships, and SVM provides robustness prediction across various feature sets. However, NN is sensitive to irrelevant input data and requires careful parameter tuning, and SVM is computationally intensive and less accurate at high wind speeds.

Gu et al. [10] have utilized the RF to forecast the wind power at a coastal wind farm in Zhejiang. Their method has shown promising performance, in particular for high wind speed compared to another linear model. However, the method drops its accuracy at low wind speeds and depends on the feature. These studies are limited by their computational complexity, requirement for parameter tuning, and dependence on feature selection for efficient results. Despite these drawbacks, traditional machine learning models remain adequate for wind power forecasting.

2.2.2 Deep Learning Models

Subsequently, a new subset of ML was introduced, known as deep learnin (DL). What makes deep learning models unique is that they can handle extensive data, deal with sequential data perfectly, and find patterns from non-linear relationships between features. DL has been used in wind power prediction, such as CNN, RNN, and their variants, LSTM and GRU. Yu et al. [34] have proposed a method CNN that converts the wind turbines into a spatial grid, giving CNN an advantage because it can extract spatiotemporal features. Hence, their proposed method reduces the prediction error by half and less time consumption than traditional time series models. Furthermore, Moharm et al. [22] have studied LSTM and Bi-LSTM to predict the wind speed at Gabal El-Zayt Wind Farm in Egypt. Their experiment result is that the SoftSign function for the state activation function and Sigmoid for the gate activation function have enhanced the prediction error result, particularly for the RMSE metric, compared to other experiments of different activation functions.

Ding et al. [7] have utilized the GRU model for wind speed error correction, which integrates Bi-Directional GRU(Bi-GRU) with a power curve model. Therefore, they corrected the NWP wind speed errors using their method, which outperformed both SVM and ANN. Furthermore, this study [25] compare varying deep learning models such as LSTM, Bi-LSTM, GRU, and BiGRU, in sequential data forecasting for different applications; the results show that Bi-LSTM often outperforms alternatives. Zhang et al. [35] have proposed a hybrid model of both CNN and LSTM. This hybrid model gets the benefits of each model. In this case, CNN is used to extract features, while LSTM is used for sequential forecasting. The features fed to the hybrid model were the meteorological features, improving predictive accuracy over conventional statistical models. However, these methods counter issues such as computational cost and hyperparameter selection and require large and high-quality datasets. Tackling these issues is important to further achieving robustness and accurate forecasting for deep learning models in the field of wind power forecasting.

Transformers were introduced in 2017 by the Google research team [30]. Their methodology has been applied to various fields, such as natural language processing (NLP), which led many companies to adopt the transformer architecture to improve the performance of large language models, including GPT [26], LLAMA [29], and others. Transformers have also been introduced to the time series in different fields, such as financial markets, weather, etc. This study [6] has explored the use of Transformers and another model for wind power prediction over 15-day horizons over two datasets. They evaluated model performance using R^2 scores, MAE, and RMSE. The results showed that the Transformer achieved better outcomes than alternative methods. However, when the paper combined the LSTM and the transformer by taking advantage of both models, it achieved more significant results than the individual models.

2.3 Decompositions Approaches

Researchers have focused on using decomposition techniques such as Wavelet Decomposition (WD) and Empirical Mode Decomposition (EMD) with prediction methods to handle the issue of wind power data. Those techniques effectively handle non-stationary data by separating the original signal into multiple stable sub-series. The WD technique analyzes time series by subdividing into frequency components using a series of high and low passes [23]. Its capabilities in the multi-resolution analysis in time and frequency domains that is effective in handling non-stationary signals. Based on Figure 2.1, the WD breaks the signal into low and high frequency components, so the WD continues to decompose the low at each level to obtain one lower frequency approximate subseries and several higher frequency detail subseries. The advantage of using the WD is the ability to capture high-frequency and low-frequency information containing approximation and significantly improve prediction by combining several models [23]. On the other hand, the disadvantage is the non-adaptive nature that demands careful selection of wavelet types and levels of decomposition.

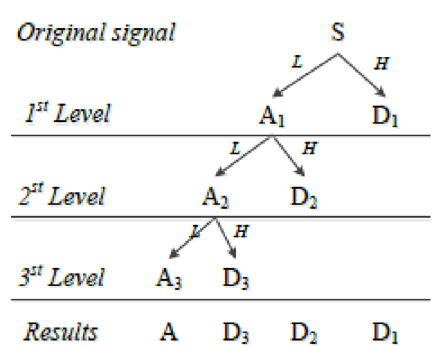

Figure 2.1: Wavelet Decomposition [23].

The EMD is an adaptive technique built to handle non-linear and non-stationary time series by decomposing the original signal into Intrinsic Mode Functions (IMFs) [23]. The

EMD has been widely used for hybrid forecasting models by enhancing prediction accuracy since it effectively deals with nonstationarityy of time series data [23]. The advantage of the EMD is its adaptive nature to capturing local signal features. However, its disadvantages include the mode mixing problem, meaning that a signal of different scales can exist in one IMF or a signal of similar scales resides in different IMFs [32].

By developing forecasting models for each of these subsequences, these methods significantly improve both accuracy and robustness in recognizing complex patterns. Integration of decomposition with deep learning models has gained attention in the field of wind power forecasting. [38] have proposed a hybrid model that utilizes both the EMD and Radial Basis Function Neural Network (RBFNN), where the EMD tasks are to break down the wind power signal into intrinsic mode functions (IMFs). Then, each IMF was fed into the RBFNN. Their proposed method has enhanced prediction accuracy compared to traditional ANN models, but requires a focus on selecting the decomposition parameters. [23] have reviewed different types of hybrid models based on decomposition, in which they have discussed WD, EMD methods in the context of wind forecasting. They have analyzed the influences of using the decomposition, which enhances the accuracy of the prediction.

These studies have shown that integrating the decomposition technique with the model will enhance prediction by breaking down the nonstationary time series into more stable components. Decomposition techniques provide benefits for researchers by improving the prediction accuracy of models by understanding complex patterns. However, the wind power data contains significant noise that requires the use of an advanced decomposition method to handle this issue. Advanced decomposition, such as CEEMDAN and EWT, offers benefits for dealing with nonstationary, nonlinear, and noise from wind power data. These methods produce a high input quality by better feature extraction and noise reduction to enhance the prediction accuracy of models. By combining CEEMDAN, EWT, and GRU, these methods improve the accuracy and robustness of wind power forecasting.

2.4 Summary

This chapter summarized the related work in the field of wind power forecasting. Although each category has its advantages, but it contains limitations that impact their prediction accuracy. For physical modeling like the NWP, which heavily depends on the meteorological feature, discarding the historical wind data. Statistical methods such as AR and ARIMA are well performed for linear time series forecasting but fail to deal with nonlinear relationships from wind power data. Their limitations influence their prediction accuracy by getting less accurate in volatile situations.

Deep learning models, such as RNN, LSTM, and transformers, have effectively handled complex non-linear relationships but remain susceptible to noisy and non-stationary data. Since the nature of the wind is difficult to predict, it can introduce inconsistent learning patterns, causing performance degradation in deep learning-based forecasting models. Furthermore, the majority of research papers did not create an extensive experiment to evaluate the influence of input and output configurations on forecasting methods. Selecting the optimal input and output configurations is important. So, there is a gap in existing research regarding a detailed analysis of how these configurations impact predictive accuracy and computational efficiency.

Another research gap in the literature is the limited use of the advanced decomposition method with deep learning for wind power forecasting. Several papers have utilized decomposition methods such as WD and EMD, which effectively enhance prediction accuracy. Utilizing the CEEMDAN and EWT with deep learning models remains inadequate. These advanced decompositions offer promising handling of the nonstationary, nonlinear, and noisy wind data by decomposing the original signal into more stable components. These provide an opportunity to explore and utilize this method with deep learning models. Tackling these challenges is an important task to enhance the prediction accuracy for wind power forecasting.

Chapter 3

Methodology

As discussed in Chapter 2, there are gaps in utilizing the advanced method with deep learning models. This chapter introduces a new methodology that integrates three components. The first component, Complete Ensemble Mode Decomposition with Adaptive Noise (CEEMDAN), is detailed in Section 3.1, Section 3.2 outlines the Empirical Wavelet Transform (EWT), Section 3.3 discusses the specifics of the Gated Recurrent Unit (GRU) approach. Lastly, Section 3.4 provides a step-by-step description of the framework and how the internal components interact.

3.1 Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN)

Wind power exhibits a non-linear and non-stationary nature, which will challenge the model. However, Empirical Mode Decomposition (EMD), proposed by [14], which this method subdivides the signals that contain nonlinearity and nonstationarity into a series of Intrinsic Mode Functions (IMFs) and a residual with a variety of frequencies based on local properties of the time series adaptively and efficiently. Furthermore, EMD has demonstrated its effectiveness for time series tasks containing both non-linearity and stationary cases. Unfortunately, EMD has its limitations, one of which is a mode mixing problem. This means that signals of different scaling frequency components can overlap across various IMFs. To address the main drawback of mode mixing, [32] has to introduce a variant of EMD called Ensemble Empirical Mode Decomposition (EEMD). This proposed method adds white noise to the original signal before applying the EMD onto it and averages the resulting IMFS to mitigate the mode mixing issue. However, EEMD provides residual noise and high computational cost.

Furthermore, CEEMDAN [28] has been introduced to address the limitations of EEMD by ensuring that each IMF is extracted using adaptive noise at each stage, leading to a complete noise-free decomposition by which each IMF represents a distinct frequency component. Decomposing the original signal into IMFs ensures reducing noise and capturing local features to produce high-quality input for forecasting. The detailed steps are as follows [17]:

1. **Add noise to the original signal:**Build various realizations of the original signal by adding adaptive Gaussian white noise:

$$x_i(t) = x(t) + \epsilon_0 w_i(t), \quad i = 1, 2, \ldots, L \tag{3.1}$$

where ϵ_0 is the noise amplitude, $w_i(t)$ is the i-th realization of white noise, and L is the total number of realizations.

2. **Extract the first IMF:** Apply EMD [14] for each realization to get the first IMF and then compute average of the first IMF:

$$I\tilde{M}F_1(t) = \frac{1}{L} \sum_{i=1}^{L} IMF_{1,i}(t) \tag{3.2}$$

where $IMF_{1,i}(t)$ is the first IMF of the i-th noisy signal.

3. **Compute the first residual:** Subtract the first IMF from the original signal to obtain the first residual:

$$r_1(t) = x(t) - I\tilde{M}F_1(t) \tag{3.3}$$

4. **Iteratively extract subsequent IMFs:** For $k \geq 2$, construct multiple realization of the residual by adding adaptive noise to the residual, apply EMD to each realization and compute the average to extract the k-th IMF:

$$I\tilde{M}F_k(t) = \frac{1}{L} \sum_{i=1}^{L} EMD\left(r_{k-1}(t) + \epsilon_k E_k(w_i(t))\right) \tag{3.4}$$

where ϵ_k is the noise scaling coefficient and $E_k(\cdot)$ represents the k-th mode extracted using EMD.

5. **Update the residual:** After extracting the k-th IMF, update the residual by subtracting the k-th IMF from the previous residual :

$$r_k(t) = r_{k-1}(t) - I\tilde{M}F_k(t) \tag{3.5}$$

6. **Stop criterion:** Repeat steps 4 and 5 until the residual $r_k(t)$ becomes a monotonic function that EMD cannot further decompose.

Figure3.1 shows the original wind power data and how it is decomposed into several IMFs shown in figure 3.2.

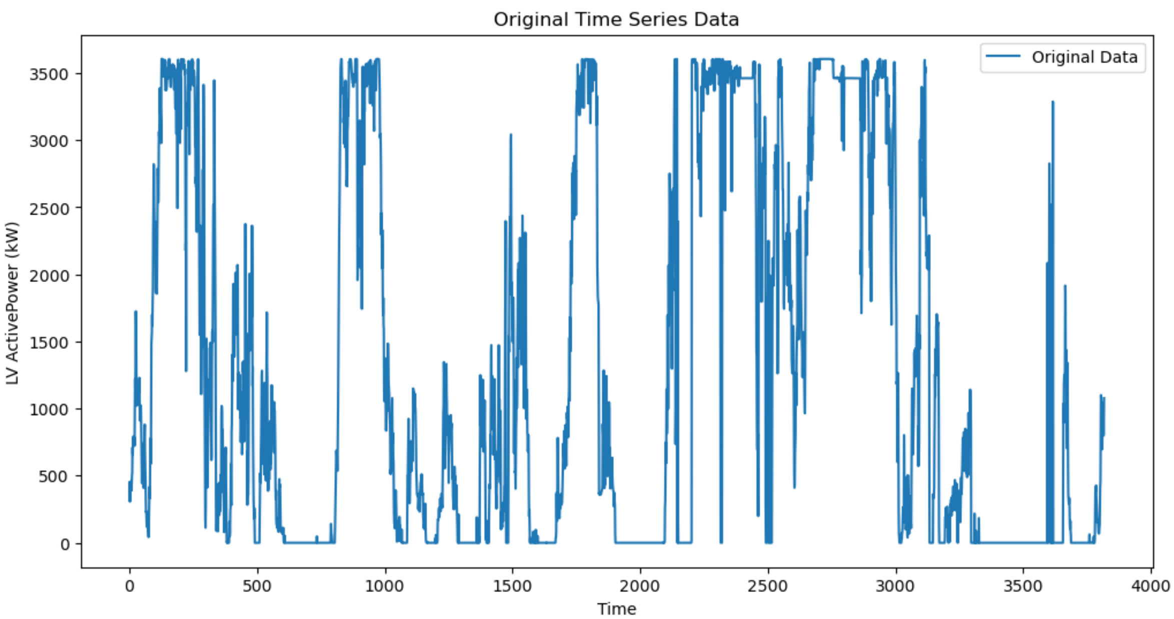

Figure 3.1: Wind Power Data in January.

3.2 Empirical Wavelet Transform (EWT) for High-Frequency Refinement

As discussed in the previous sub-section, CEEMDAN effectively decomposes the signal or time series into IMFs. However, the issue is that the first IMF obtains high-frequency components, which include noise. To further solve this issue, we utilize the Empirical Wavelet Transform (EWT) introduce by [9]. This method guarantees that the meaningful feature of the first IMF is preserved while the noise is minimized. Compared to traditional wavelet transforms, the EWT is an adaptive signal decomposition technique that constructs empirical scaling and wavelet functions based on the Fourier spectrum of the signal. The main goal of EWT is to segment the Fourier domain and build wavelet filters that adaptively isolate different frequency components. This ensures that the EWT is useful for different fields that involve the application of time series since dealing with varying frequency characteristics improves the signal clarity before being fed to the model for forecast purposes.

The EWT algorithm comprises five main steps. The first one is to apply the Fourier Transform to the original signal $x(t)$. Second, it builds a frequency boundaries Ω_i by detecting the local maxima of the frequency spectrum, before defining boundaries that partition the dominant frequency to obtain the frequency boundaries. Third, it constructs the scaling function $\hat{\phi}_n(\omega)$ and the wavelet function $\hat{\psi}_n(\omega)$ for each frequency segment. Fourth, after obtaining wavelet filters from different frequency segments, the wavelet coefficients $W_E^{(n)}(t)$ are obtained by convolving the signal with the wavelet functions. Similarly, the approximation coefficients $W_E^{(0)}(t)$ are obtained using the scaling function. The last step is that the original signal $x'(t)$ is reconstructed from the wavelet coefficients. Detailed steps

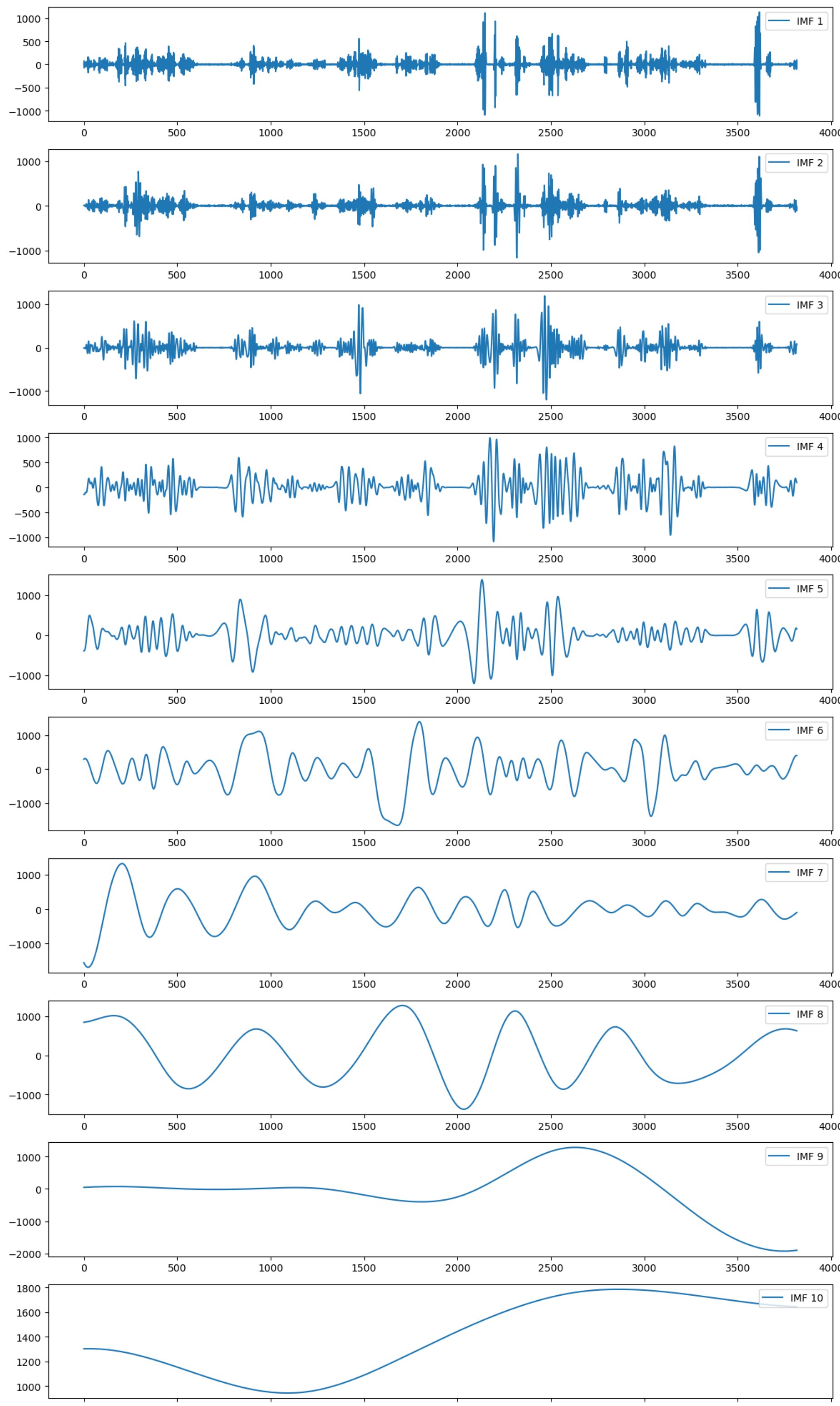

Figure 3.2: Decomposed IMFs of Wind Power Data for January.

are as follows [17]:

1. **Compute the Fourier Transform:** Convert a signal from time domain into frequency domain using Fourier Transform:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \tag{3.6}$$

   where $X(\omega)$ represents the signal in the frequency domain.

2. **Identify frequency boundaries:** Identify peaks in the Fourier spectrum and divide the frequency domain into different segments by defining boundaries based on those maxima:

$$\Omega_i = \frac{\omega_i + \omega_{i+1}}{2}, \quad i = 1, 2, \ldots, P-1 \tag{3.7}$$

   where ω_i represents the identified dominant frequencies.

3. **Construct adaptive wavelet filters:** Construct the empirical wavelet bank filters for each segmented Fourier spectrum.

   - **Scaling Function (Low-Pass Filter):**

$$\hat{\phi}_n(\omega) = \begin{cases} 1, & |\omega| \leq (1-\gamma)\omega_n \\ \cos\left(\frac{\pi}{2}\beta\left(\frac{|\omega|-(1-\gamma)\omega_n}{2\gamma\omega_n}\right)\right), & (1-\gamma)\omega_n \leq |\omega| \leq (1+\gamma)\omega_n \\ 0, & \text{otherwise} \end{cases} \tag{3.8}$$

   - **Wavelet Function (Band-Pass Filter):**

$$\hat{\psi}_n(\omega) = \begin{cases} 1, & (1+\gamma)\omega_n \leq |\omega| \leq (1-\gamma)\omega_{n+1} \\ \cos\left(\frac{\pi}{2}\beta\left(\frac{|\omega|-(1-\gamma)\omega_{n+1}}{2\gamma\omega_{n+1}}\right)\right), & (1-\gamma)\omega_{n+1} \leq |\omega| \leq (1+\gamma)\omega_{n+1} \\ \sin\left(\frac{\pi}{2}\beta\left(\frac{|\omega|-(1-\gamma)\omega_n}{2\gamma\omega_n}\right)\right), & (1-\gamma)\omega_n \leq |\omega| \leq (1+\gamma)\omega_n \\ 0, & \text{otherwise} \end{cases} \tag{3.9}$$

4. **Perform the EWT:** After obtaining empirical wavelet bank filters, apply them to extract frequency components from the first IMF. The wavelet coefficients are obtained by convolving the first IMF with the wavelet functions:

$$W_E^{(n)}(t) = \int_{-\infty}^{\infty} x(\tau)\psi_n(\tau - t)d\tau \tag{3.10}$$

   Similarly, the approximation coefficients using the scaling function are:

$$W_E^{(0)}(t) = \int_{-\infty}^{\infty} x(\tau)\phi_1(\tau - t)d\tau \tag{3.11}$$

5. **Reconstruct the signal:** The first IMF is reconstructed by preserving important information and reducing noise using the wavelet coefficients:

$$x(t) = W_E^{(0)}(t) * \phi_1(t) + \sum_{n=1}^{P} W_E^{(n)}(t) * \psi_n(t) \tag{3.12}$$

where $*$ denotes convolution.

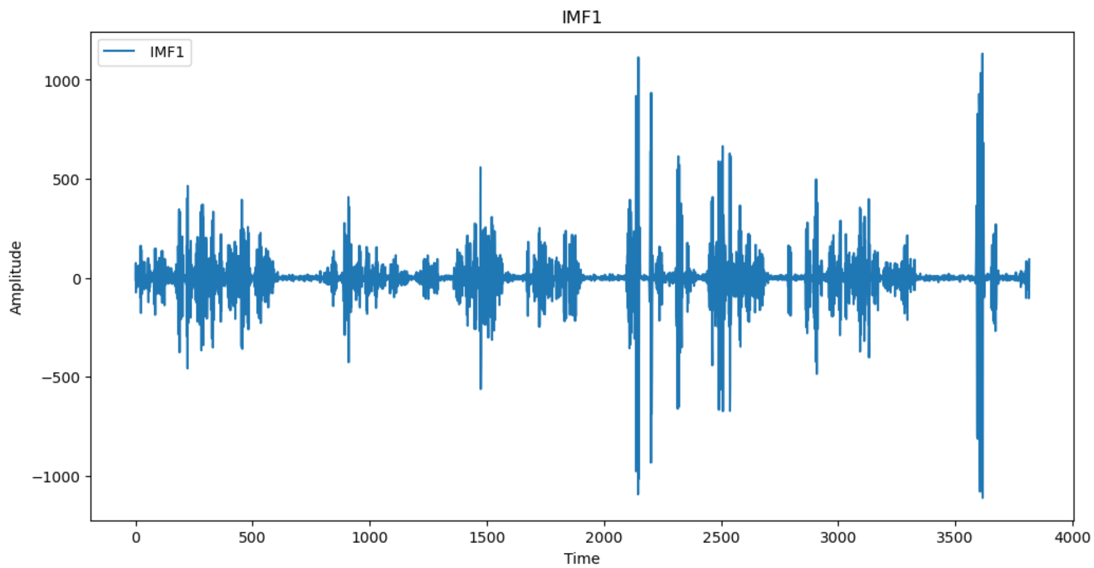

Figure 3.3: The First IMF.

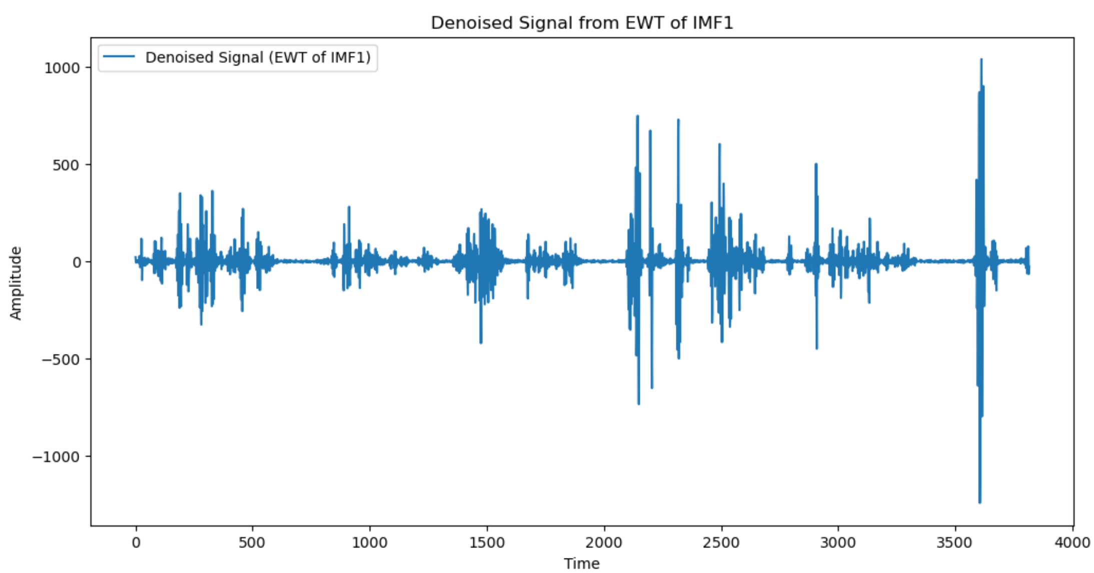

Figure 3.4: Denoised Signal of IMF1 Using EWT

As shown in Figure 3.3, the first IMF contains high-frequency components with noise. The noise is minimized after applying EWT to this IMF, as demonstrated in Figure 3.4.

3.3 Gated Recurrent Unit (GRU)

The GRU has been introduced as a variant of RNN and LSTM, with several known improvements. This model can capture the dependence relationship for the short and long-term time series and also solve the vanishing gradient problem, which has been a limitation of the RNN [37]. Furthermore, the GRU consists of two gates that are much less than the LSTM and are simple: the update and reset gates.

The update gate's purpose is to control the amount of the prior gating structure retained in the current unit, while the reset gate's purpose is to determine whether the information from the past and present time steps can be closely combined. Hence, its simple structure reduces the training time without dropping notable prediction performance, allowing it to learn for both short—and long-term intervals and making it ideal for time series forecasting. In addition, the GRU has demonstrated remarkable adaptability when handling high-frequency precipitation data in time series [37]. Hence, based on the benefits of the GRU compared to another model, we selected the GRU framework to improve prediction accuracy to process the IMFs after CEEMDAN decomposition and EWT (only for the first IMF) in this study.

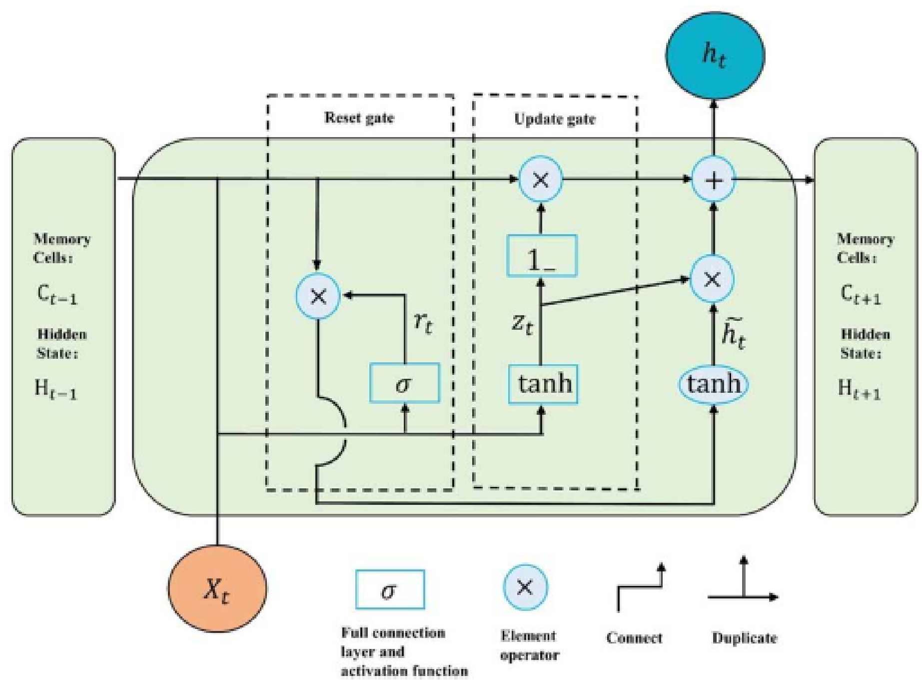

Figure 3.5: GRU architecture [37].

This Figure 3.5 shows the GRU architecture. The arrow indicates the direction of the data flow, where σ is the activation function Sigmoid, tanh is the hyperbolic tangent of the activation function, z_t and r_t represent the update and reset gates. The GRU unit calculation process is as follows:

1. **Compute the Update Gate:** The update gate controls how much the past information is going forward.

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \tag{3.13}$$

where:

- x_t is the input vector at time t.
- h_{t-1} is the hidden state from the previous time step.
- W_z and U_z are learnable weight matrices.
- b_z is the bias term.
- $\sigma(\cdot)$ is the sigmoid activation function.

2. **Compute the Reset Gate:** The reset gate controls how to combine the new input with previous memory.

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \tag{3.14}$$

where:

- W_r and U_r are weight matrices for the reset gate.
- b_r is the bias term.

3. **Compute the Candidate Hidden State:** This represents the update of the hidden state based on the current input.

$$\tilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h) \tag{3.15}$$

where:

- $\odot$ represents element-wise multiplication (Hadamard product).
- W_h and U_h are weight matrices for candidate computation.
- b_h is the bias term.
- $\tanh(\cdot)$ is the hyperbolic tangent activation function.

4. **Compute the Final Hidden State:** The final hidden state is formed by merging the previous state and candidate activation.

$$h_t = (1 - z_t) \odot \tilde{h}_t + z_t \odot h_{t-1} \tag{3.16}$$

where:

- $(1 - z_t)$ controls the influence of the new candidate $\tilde{h}_t$.
- z_t determines how much of the past hidden state h_{t-1} is retained.

3.4 Framework of the Proposed Method

As discussed in the previous sections, each component was explained individually. This section integrates all the components CEEMDAN, EWT, GRU. The framework of this proposed method follows four steps as shown in 3.6:

- **Step 1:** The original wind power data is first fed into the CEEMDAN to decompose the original data into subseries known as IMFs. Each IMF captures distinct frequencies and removes noise from the data.
- **Step 2:** As CEEMDAN decomposes the original data into several IMFs, the first IMF contains high-frequency noise. Then, it is further processed using EWT. These steps refine the first IMF by preserving meaningful features while reducing noise.
- **Step 3:** GRU is used for forecasting purposes by predicting all the IMFs obtained from the CEEMDAN-EWT.
- **Step 4:** All forecast results for each IMF are added together to obtain the final forecast results.

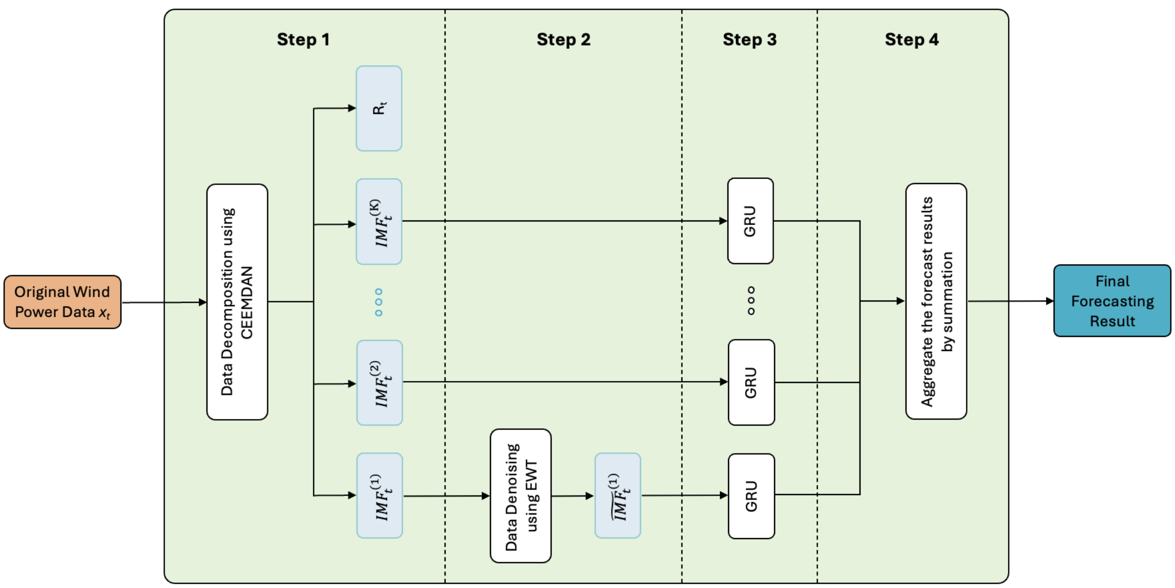

Figure 3.6: Framework of the Proposed Method.

The following steps outline the mathematical formulation of the framework.

Step 1: Decomposing the Feature Using CEEMDAN

The feature x_t is decomposed using Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) to obtain K Intrinsic Mode Functions (IMFs):

$$x_t \xrightarrow{\text{CEEMDAN}} \{IMF_t^{(1)}, IMF_t^{(2)}, \ldots, IMF_t^{(K)}\} \tag{3.17}$$

where:

- x_t is the observed value of the feature at time t.
- $IMF_t^{(k)}$ represents the k-th IMF of the single feature at time t.

Step 2: Processing IMF1 Using EWT

The first IMF $IMF_t^{(1)}$ is further processed using Empirical Wavelet Transform (EWT):

$$IMF_t^{(1)} \xrightarrow{\text{EWT}} I\tilde{M}F_t^{(1)} \tag{3.18}$$

where $I\tilde{M}F_t^{(1)}$ is the transformed version of $IMF_t^{(1)}$ after EWT processing.

Step 3: Training GRU for Each IMF Independently

Each IMF $IMF_t^{(k)}$ is used as an input to a separate GRU model:

$$\hat{y}_t^{(k)} = f_{\text{GRU}}^{(k)}(IMF_t^{(k)}) \tag{3.19}$$

where:

- $\hat{y}_t^{(k)}$ is the prediction at time t for IMF k.
- $f_{\text{GRU}}^{(k)}(\cdot)$ is the GRU model trained for the k-th IMF.

Step 4: Final Prediction Aggregation

The final output is obtained by summing the predictions from all GRU models:

$$\hat{y}_t = \sum_{k=1}^{K} f_{\text{GRU}}^{(k)}(IMF_t^{(k)}) \tag{3.20}$$

where:

- $\hat{y}_t$ is the final predicted value at time t.
- K is the total number of IMFs.

3.5 Summary

This chapter provides details on the various components used in the proposed methodology for wind power forecasting and then introduces the framework of the proposed methodology. The CEEMDAN decomposes the original data into several IMFs to address the issue of non-stationarity. EWT refines the high-frequency noise of the first IMF before it is forecasted by the model. The GRU has been selected for the model to capture temporal dependencies and generate predictions with better accuracy. The framework presents all the necessary steps from decomposition to forecasts.

The next chapter details the experimental setup and evaluation framework, including dataset characteristics, hyperparameter selection, and different experiments with each discussion and analysis.

Chapter 4

Experiments, Results, and Analysis

This chapter aims to demonstrate the capabilities of the proposed method for wind power forecasting. Section 4.1 will discuss the dataset used for evaluation and the preprocessing steps. Furthermore, Section 4.2 focuses on the model training hyperparameter used in the experiments. Section 4.3 describes the evaluation metrics used in the experiments. In this study, three experiments are conducted to ensure a comprehensive evaluation. Each experiment is described in detail in Section 4.4, Section 4.5, and Section 4.6. Finally, Section 4.7 discusses and analyzes the experimental outcomes.

4.1 Dataset and Preprocessing

This section explains in detail the dataset that is being used and the pre-processing steps that have been performed.

4.1.1 Dataset

The dataset used in this study for wind power forecasting is collected in Turkey, where it measures wind speed, wind direction, theoretical power curves, and actual power output for 10-minute intervals for the whole year of 2018, [15].

The features that are included in this dataset are **LV ActivePower**, which shows the current power output(KW); **Wind Speed** at the turbine height (m/s) to generate the electricity; **Wind Direction (in degrees)**, which is the wind direction at the hub height of the turbine; **Theoretical power** is the ideal power output of the wind turbine based on the turbine's specifications and wind speed, this feature is derived from the turbine power curve that standardized the relationship between the wind speed and power output. Figure 4.1 below, is the wind power data for each month.

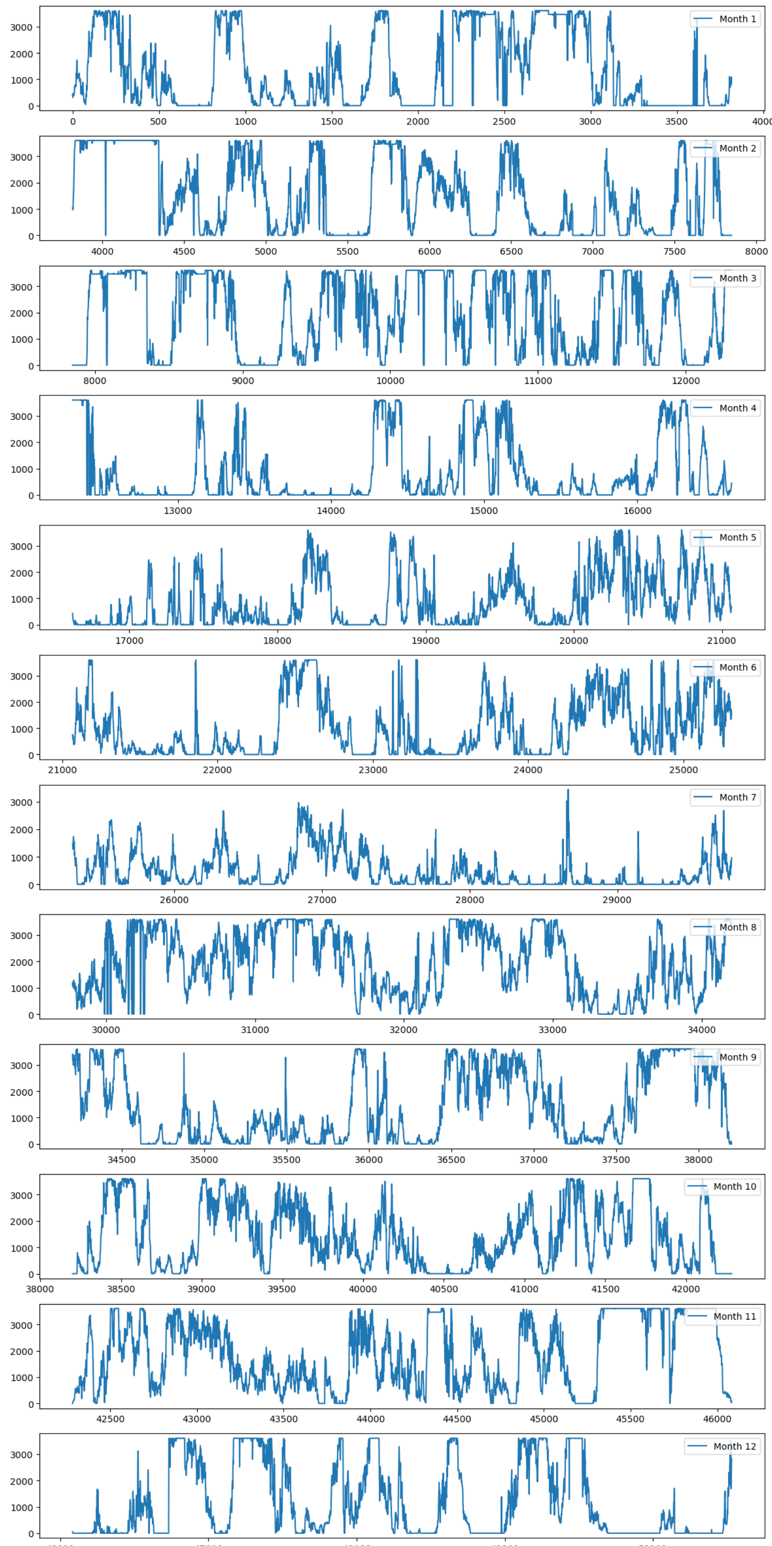

Figure 4.1: Monthly Time Series Plots of Wind Power Generation.

4.1.2 Data Preprocessing

This section explains the preprocessing steps followed in the code. In most research studies, the first step in preprocessing is to check if the data contains missing values. In this study, no missing values were found. Therefore, there is no need for a filling strategy. The proposed method is evaluated monthly, as shown in Figure 4.1. Therefore, the data is split by months. Next, CEEMDAN and EWT are applied as described in Sections 3.1 and 3.2. Since the GRU model is trained for each IMF, all IMFs are divided into training and test sets using an 80:20 split.

In time series forecasting, the data is structured as input-output pairs. The training and test sets consist of these pairs to train and evaluate the proposed method effectively. For example, if six data points (representing a 1-hour duration) are used to predict the next 10-minute value, each input sequence will contain six points, with a single data point as output.

MinMax Scaler:

$$X_{\text{scaled}} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \tag{4.1}$$

Standard Scaler:

$$X_{\text{standardized}} = \frac{X - \mu}{\sigma} \tag{4.2}$$

where:

- μ is the mean of the dataset.
- σ is the standard deviation.

After, the standard scaling was applied to all the training and test elements. Then, the scaling of the prediction is inversed to evaluate the performance of the proposed method based on evaluation metrics. Those steps have been applied to the first and third experiments.The second experiment differs slightly in that the MinMaxScalar is applied before decomposing the data. Also, for the wind direction to follow its nature, we applied the 1/(360-0) came from the MinMaxScalar, where the $Xmin$ is equal to zero and $Xmax$ is equal to 360 degrees. After scaling, the CEEMDAN and EWT are applied to all features. Then, it follows the same steps as the previous experiments. StandardScalar has introduced a downward shift in prediction when adding more features to the proposed method since the StandardScalar main focus is to scale the data around the mean zero, which introduces negative and positive values; this has caused downward shifts. On the other hand, MinMaxScalar preserves the relative distributions of the feature but shifts the data into the range [0,1].

Different forecasting horizons exist for the input and output configurations, such as short, medium, and long-term. In this particular study, short-term forecasting was selected. For the input lengths, the (1 hour, 1.5 hours, 2 hours, and 4 hours) were selected, and the different forecasting horizons are (10 min, 20 min, 30 min, and 1 hour). This study has experimented with the performance of methods for different combinations of input and forecasting horizons, a total of 16 combinations, which will be shown in the next section 4.4.

4.2 Model Training and Hyperparameters

The hyperparameters of this study, are Adam for the optimizer with a learning rate of 0.001, the number of units 128 with a batch size of 64, the proposed method has been trained for 30 epochs, and the loss function is MSE. The packages for the proposed method are pyEMD for the CEEMDAN decomposition [19], and the EWT has been taken from the ewtpy package [4]. Building the GRU is done using the Keras and Tensorflow libraries. Those configurations for our experiments.

4.3 Evaluation Metrics

To evaluate the performance of the proposed method with other methods, we have utilize several evaluation metrics. These metrics demonstrate the accuracy, reliability, and efficiency by comparing between the prediction with actual wind power values. The selected metrics include Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-squared (R^2) score, and Margin of Error, which are widely used in time-series forecasting.

4.3.1 Root Mean Squared Error (RMSE)

RMSE is a commonly used metric that measures the average magnitude of the forecasting error, giving higher weight to large errors due to the squared term. It is calculated as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4.3}$$

where:

- y_i is the actual wind power value.
- $\hat{y}_i$ is the predicted wind power value.
- n is the total number of data points.

4.3.2 Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values, providing an interpretable measure of forecasting accuracy. It is given by:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4.4}$$

4.3.3 R-Squared ($\mathbf{R^2}$) Score

The R^2 score, or the coefficient of determination, evaluates how well the model explains the variance in the actual wind power data. It is computed as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{4.5}$$

where:

- y_i is the actual wind power value.
- $\hat{y}_i$ is the predicted wind power value.
- $\bar{y}$ is the mean of the actual wind power values.
- n is the total number of data points.

4.3.4 Margin of Error

Margin of Error is a statistical term that represents the range of uncertainty around an estimate. It is computed as:

$$E = Z \times \frac{\sigma}{\sqrt{n}} \tag{4.6}$$

where:

- Z is the z-score corresponding to the desired confidence level (e.g., 1.96 for 95% confidence).
- σ is the standard deviation of the sample errors.
- n is the total number of observations (sample size).

4.4 Experiment 1: Benchmarking the Proposed Model Against Baseline Models

This experiment aims to evaluate the performance of the proposed method CEEMDAN-EWT-GRU framework by comparing it with baseline methods such as RNN, LSTM, GRU, Bi-LSTM and Transformer, benchmark (CEEMDAN-EWT-LSTM) [17], and hybrid model like CEEMDAN-EWT-Transformer for varying input-output lengths.

4.4.1 Experimental Setup

In this experiment, we tested the models for input lengths of 1 h, 1.5 h, 2 h and 4 h. Regarding the forecasting horizon, we chose 10 min, 20 min, 30 min and 1 hr. Therefore, in total, we performed 16 tests for the model's performance to see how consistent the performance of the proposed method is compared to others.

4.4.2 Results and Discussion

Table 4.1 demonstrates the performance of the models from baselines, benchmarks, hybrid models, and proposed method for an input length of 1 hour and forecasting of 10 min, which is the setting of the benchmark paper [17]. Furthermore, we also evaluate the performance of the proposed method, the benchmark model, and the hybrid models that integrate both CEEMDAN and EWT. This comparison highlights the effectiveness of the decomposition techniques in enhancing forecasting performance. Our goal is to identify the best-performing methods. It appears that the proposed method has outperformed all other methods for 11 months, with a total mean of RMSE (92.053), MAE (59.831), and R2 score (0.989), which illustrates how effective it is in prediction accuracy. Additionally, there is an insight that the proposed method scores an RMSE higher than 100 from January to June, suggesting the need for an ensemble method. From July to December, the proposed method achieves an RMSE lower than 100, indicating that using a single model would be sufficient, thereby reducing computational costs.

On the other hand, from the baselines model, both the Bi-LSTM and Transformer are the best-performing models compared to other baselines. However, when both of them were integrated with CEEMDAN and EWT, they did not perform as well as the proposed method. There are several reasons for this case. In the context of the CEEMDAN-EWT-Transformer model, the paper [23] has highlighted the advantages and disadvantages of EMD and its variants. CEEMDAN offers advantages such as adaptability, resolving the mode mixing issue present in EMD, and preventing the introduction of extra noise in the reconstructed signal. However, a drawback of CEEMDAN is its sequential nature, which prevents parallel processing. This poses a challenge for the Transformer, as its efficiency relies on parallel computation, particularly during training and inference. The sequential processing of CEEMDAN creates a bottleneck, limiting the advantages of the Transformer's parallelized attention mechanisms and matrix operations.

Furthermore, the CEEMDAN-EWT-GRU model outperforms other hybrid models, such as CEEMDAN-EWT-BiLSTM and the benchmark model (CEEMDAN-EWT-LSTM), which is consistent with the findings of Zhao et al. [37]. The benefits of processing the decomposed signals effectively compared to both the LSTM and Bi-LSTM, which have colossal parameters to train compared to GRU, which takes less training time due to its simple architecture and fewer parameters, since LSTM have more gates compared to the GRU this additional increases the complexity which is more advantageous for the case if raw/ original data, but dealing with CEEMDAN and EWT the burden of capturing long-term dependencies was already alleviated by decomposition, making LSTM's complexity

Model	Metrics	Jan	Feb	March	April	May	June	July	August	Sept	Oct	Nov	Dec	Average	Sum
Transformer	RSME	303.769	337.06	337.77	275.645	383.119	364.777	133.419	329.653	327.721	254.302	330.843	144.958	293.586	3523.036
	MAE	169.387	215.915	228.503	203.776	287.115	254.79	78.333	284.386	270.494	205.926	283.193	110.23	216.004	2592.048
	R2	0.818	0.924	0.945	0.948	0.828	0.862	0.929	0.913	0.932	0.96	0.884	0.898	0.903	10.841
RNN	RSME	306.116	309.844	340.632	182.063	378.39	387.04	146.304	433.305	277.605	660.569	255.5	193.325	322.558	3870.693
	MAE	125.373	237.603	262.925	122.658	281.454	280.922	75.711	388.814	217.617	493.573	201.612	177.783	238.837	2866.045
	R2	0.815	0.935	0.944	0.977	0.833	0.844	0.915	0.85	0.951	0.729	0.931	0.818	0.879	10.542
LSTM	RSME	284.606	321.078	523.171	224.114	437.65	390.345	143.354	536.128	424.752	469.072	317.835	187.92	355.002	4260.025
	MAE	162.914	238.959	448.153	165.147	330.957	284.547	77.384	422.277	355.027	361.581	205.464	163.88	268.024	3216.29
	R2	0.84	0.931	0.867	0.965	0.776	0.841	0.918	0.77	0.886	0.863	0.893	0.828	0.865	10.378
GRU	RSME	296.017	315.61	410.94	208.442	700.199	445.345	141.179	390.569	354.488	420.751	243.922	197.536	343.75	4124.998
	MAE	188.871	241.312	357.801	161.729	564.648	335.714	80.204	341.732	290.468	327.232	175.378	182.364	270.621	3247.453
	R2	0.827	0.933	0.918	0.97	0.427	0.794	0.921	0.878	0.921	0.89	0.937	0.81	0.852	10.226
Bi-LSTM	RSME	278.346	272.156	289.604	187.959	336.597	362.967	128.563	247.936	235.54	257.309	221.25	112.407	244.22	2930.634
	MAE	101.444	138.946	181.576	115.828	244.266	247.79	70.851	188.97	144.82	191.047	137.975	62.901	152.201	1826.414
	R2	0.847	0.95	0.959	0.976	0.868	0.863	0.934	0.951	0.965	0.959	0.948	0.938	0.93	11.158
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**159.848**	**104.645**	**115.854**	**65.954**	**120.018**	**129.677**	**44.944**	**80.535**	**96.789**	**64.776**	85.281	**36.311**	**92.053**	**1104.632**
	MAE	**113.258**	**62.112**	**67.037**	**41.752**	**81.743**	**94.476**	**25.51**	**54.866**	**70.223**	**45.104**	40.109	**21.787**	**59.831**	**717.977**
	R2	**0.95**	**0.993**	**0.993**	**0.997**	**0.983**	**0.983**	**0.992**	**0.995**	**0.994**	**0.997**	0.992	**0.994**	**0.989**	**11.863**
CEEMDAN-EWT-LSTM (Benchmark)	RSME	202.458	108.26	121.099	67.965	124.621	132.089	45.811	81.868	151.252	72.272	**84.338**	36.726	102.397	1228.759
	MAE	146.862	68.415	70.416	42.866	84.172	97.61	26.578	56.907	120.283	52.647	**42.483**	22.549	69.316	831.788
	R2	0.919	0.992	0.993	0.997	0.982	0.982	0.992	0.995	0.986	0.997	**0.992**	0.993	0.985	11.82
CEEMDAN-EWT-BiLSTM	RSME	185.342	115.757	125.428	67.587	125.268	133.228	46.435	84.128	143.651	74.643	86.547	107.076	107.924	1295.09
	MAE	127.545	73.032	76.937	44.338	84.239	97.899	28.391	61.664	114.453	55.023	41.815	81.333	73.889	886.669
	R2	0.932	0.991	0.992	0.997	0.982	0.982	0.991	0.994	0.987	0.997	0.992	0.944	0.982	11.781
CEEMDAN-EWT-Transformer	RSME	382.646	173.029	241.975	109.585	210.966	208.302	66.362	132.306	370.888	148.649	123.775	150.738	193.268	2319.221
	MAE	278.782	102.245	149.807	63.986	149.908	150.169	35.895	88.671	298.668	115.095	58.034	110.931	133.516	1602.191
	R2	0.756	0.98	0.97	0.992	0.95	0.955	0.981	0.983	0.906	0.986	0.961	0.881	0.942	11.301

Table 4.1: Forecasting Accuracy of Different Methods for 1-Hour Input Length and 10-Minute Horizon.

unnecessary. On the other hand, the GRU can focus more on temporal dependencies from decomposed data, especially in the high-frequency IMF components [37].

The second reason is that according to Zhao et al. [37], GRU is better at handling high-frequency components in decomposed data. At the same time, it is capable of maintaining computational efficiency. This is primarily due to GRU's reset gate, which allows it to dynamically ignore irreverent past information, making it more adaptive to high-frequency IMFs. However, LSTM and Bi-LSTM retain more long-term information, which is not applicable when dealing with high-frequency IMFs, leading to overfitting or redundant information [37].

Based on the performance metrics across different historical lengths and forecast horizons shown in Tables 4.2 4.3 4.4 4.5, the proposed method (CEEMDAN-EWT-GRU) consistently outperforms the other forecasting methods by achieving the lowest evaluation metrics in terms of RMSE and MAE while highest R^2 scores across all time intervals. This demonstrates how effective the GRU is with the CEEMDAN and EWT, as mentioned, the benefits of the GRU compared to others. The benchmark (CEEMDAN-EWT-LSTM) performs slightly worse than the proposed methods, but still outperforms the baseline model LSTM, GRU, and Transformer, suggesting that decomposition techniques significantly improve forecasting ability. Bi-LSTM has shown excellent performance compared to other deep learning models for different time horizons. Furthermore, the CEEMDAN-EWT-BiLSTM did outperform the benchmark in most time horizons but did not surpass the proposed method.

The CEEMDAN-EWT-Transformer has shown moderate performance, as mentioned. The drawback of the CEEMDAN with the Transformer is that their errors increase for longer forecasting horizons. However, the RNN and LSTM models exhibit the highest RMSE, MAE, and low R^2 values, suggesting that they are considered weaker predictive for wind power forecasting in this experiment. In general, the results have highlighted that the model that integrates the decomposition techniques significantly improves the accuracy of the forecasting, the proposed hybrid method based on GRU being the most effective.

4.5 Experiment 2: Impact of Additional Meteorological Features on Forecast Performance

The main objective of the second experiment is to see the performance of the proposed method CEEMDAN-EWT-GRU after adding meteorological features such as wind speed and wind direction and turbine specifications such as theoretical power curves. The selection of the historical length and forecasting horizon was based on Experiment 1, where the proposed method demonstrated the lowest RMSE and MAE, along with a high R^2 score. Consequently, a historical length of 1 hour and forecasting horizons of 10 minutes have been chosen.

Model	Metrics	1 hr - 10 mins		1.5 hr - 10 mins		2 hr - 10 mins		4 hr - 10 mins	
		Average	Sum	Average	Sum	Average	Sum	Average	Sum
Transformer	RSME	293.586	3523.036	306.499	3677.992	307.823	3693.874	300.705	3608.454
	MAE	216.004	2592.048	225.276	2703.309	241.268	2895.218	225.75	2709.004
	R2	0.903	10.841	0.889	10.665	0.89	10.674	0.891	10.697
RNN	RSME	322.558	3870.693	387.413	4648.958	298.447	3581.359	330.023	3960.27
	MAE	238.837	2866.045	305.683	3668.192	211.489	2537.868	253.346	3040.147
	R2	0.879	10.542	0.824	9.891	0.893	10.717	0.87	10.436
LSTM	RSME	355.002	4260.025	323.174	3878.088	329.386	3952.631	333.623	4003.478
	MAE	268.024	3216.29	245.355	2944.258	249.855	2998.255	257.275	3087.296
	R2	0.865	10.378	0.881	10.573	0.877	10.521	0.873	10.477
GRU	RSME	343.75	4124.998	311.541	3738.497	319.265	3831.182	315.186	3782.227
	MAE	270.621	3247.453	238.154	2857.844	239.535	2874.417	243.419	2921.032
	R2	0.852	10.226	0.889	10.662	0.881	10.569	0.88	10.559
Bi-LSTM	RSME	244.22	2930.634	239.601	2875.213	236.6	2839.204	240.979	2891.744
	MAE	152.201	1826.414	148.458	1781.491	143.16	1717.922	149.053	1788.637
	R2	0.93	11.158	0.932	11.189	0.934	11.204	0.931	11.177
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**92.053**	**1104.632**	**95.485**	**1145.823**	**94.319**	**1131.832**	**93.716**	**1124.59**
	MAE	**59.831**	**717.977**	**63.54**	**762.48**	**63.108**	**757.295**	**61.491**	**737.893**
	R2	**0.989**	**11.863**	**0.985**	**11.821**	**0.987**	**11.839**	**0.987**	**11.838**
CEEMDAN-EWT-LSTM (Benchmark)	RSME	102.397	1228.759	97.157	1165.887	112.135	1345.617	104.615	1255.377
	MAE	69.316	831.788	65.105	781.257	77.743	932.915	71.117	853.4
	R2	0.985	11.82	0.985	11.818	0.972	11.662	0.983	11.792
CEEMDAN-EWT-Bi-LSTM	RSME	107.924	1295.09	101.542	1218.5	108.461	1301.536	102.127	1225.519
	MAE	73.889	886.669	68.559	822.703	75.686	908.227	68.611	823.326
	R2	0.982	11.781	0.986	11.836	0.978	11.73	0.984	11.805
CEEMDAN-EWT-Transformer	RSME	193.268	2319.221	197.823	2373.878	202.456	2429.474	282.123	3385.481
	MAE	133.516	1602.191	134.874	1618.486	139.113	1669.359	182.674	2192.083
	R2	0.942	11.301	0.942	11.299	0.94	11.282	0.896	10.755

Table 4.2: Forecasting Accuracy of Different Methods for Varying Input Lengths with a Fixed 10-Minute Horizon.

Model	Metrics	1 hr - 20 mins		1.5 hr - 20 mins		2 hr - 20 mins		4 hr - 20 mins	
		Average	Sum	Average	Sum	Average	Sum	Average	Sum
Transformer	RSME	368.291	4419.493	330.41	3964.922	339.472	4073.669	358.223	4298.677
	MAE	273.234	3278.812	235.107	2821.278	252.063	3024.754	259.97	3119.643
	R2	0.833	9.998	0.872	10.459	0.861	10.333	0.845	10.143
RNN	RSME	419.843	5038.112	362.925	4355.099	389.01	4668.114	415.999	4991.991
	MAE	322.454	3869.447	266.953	3203.437	290.467	3485.607	323.737	3884.845
	R2	0.788	9.454	0.846	10.149	0.826	9.906	0.783	9.398
LSTM	RSME	399.286	4791.427	393.873	4726.471	367.407	4408.888	382.61	4591.323
	MAE	302.661	3631.936	300.343	3604.111	275.057	3300.685	283.157	3397.886
	R2	0.817	9.804	0.828	9.931	0.849	10.189	0.836	10.031
GRU	RSME	361.857	4342.28	355.014	4260.169	353.052	4236.627	356.191	4274.286
	MAE	270.999	3251.985	265.462	3185.541	264.791	3177.492	264.641	3175.696
	R2	0.847	10.169	0.846	10.155	0.856	10.273	0.854	10.244
Bi-LSTM	RSME	295.176	3542.108	291.242	3494.907	289.605	3475.256	288.785	3465.42
	MAE	184.851	2218.208	182.629	2191.55	178.746	2144.955	175.219	2102.623
	R2	0.898	10.779	0.901	10.808	0.902	10.824	0.903	10.832
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**124.536**	**1494.432**	**118.795**	**1425.542**	**114.656**	**1375.873**	**114.426**	**1373.115**
	MAE	**80.137**	**961.648**	**76.893**	**922.714**	**71.414**	**856.967**	**72.315**	**867.777**
	R2	**0.975**	**11.696**	**0.978**	**11.738**	**0.98**	**11.764**	**0.981**	**11.772**
CEEMDAN-EWT-LSTM (Benchmark)	RSME	134.258	1611.098	133.934	1607.206	144.748	1736.981	145.397	1744.758
	MAE	89.28	1071.362	90.319	1083.829	98.343	1180.116	98.796	1185.55
	R2	0.972	11.658	0.96	11.523	0.956	11.468	0.959	11.504
CEEMDAN-EWT-Bi-LSTM	RSME	144.397	1732.758	127.824	1533.892	130.238	1562.852	128.598	1543.171
	MAE	98.194	1178.333	85.302	1023.629	86.22	1034.639	83.277	999.327
	R2	0.961	11.529	0.972	11.666	0.969	11.632	0.976	11.713
CEEMDAN-EWT-Transformer	RSME	266.494	3197.928	272.242	3266.907	272.79	3273.479	367.161	4405.933
	MAE	194.28	2331.356	196.638	2359.654	196.451	2357.407	273.8	3285.596
	R2	0.916	10.997	0.913	10.961	0.915	10.979	0.828	9.939

Table 4.3: Forecasting Accuracy of Different Methods for Varying Input Lengths with a Fixed 20-Minute Horizon.

Model	Metrics	1 hr - 30 mins		1.5 hr - 30 mins		2 hr - 30 mins		4 hr - 30 mins	
		Average	Sum	Average	Sum	Average	Sum	Average	Sum
Transformer	RSME	387.187	4646.246	391.37	4696.445	390.979	4691.751	412.026	4944.308
	MAE	278.64	3343.684	292.054	3504.649	292.543	3510.517	308.432	3701.179
	R2	0.827	9.924	0.815	9.781	0.825	9.899	0.799	9.583
RNN	RSME	420.421	5045.046	431.75	5181.002	414.75	4976.997	443.842	5326.108
	MAE	318.516	3822.194	332.502	3990.019	311.356	3736.277	332.323	3987.88
	R2	0.792	9.508	0.782	9.378	0.804	9.648	0.768	9.21
LSTM	RSME	459.679	5516.145	432.133	5185.599	408.496	4901.947	411.389	4936.67
	MAE	354.378	4252.531	326.598	3919.176	304.631	3655.567	301.294	3615.522
	R2	0.765	9.176	0.793	9.516	0.81	9.716	0.808	9.699
GRU	RSME	393.507	4722.089	381.812	4581.741	386.333	4635.997	385.283	4623.394
	MAE	293.63	3523.556	280.973	3371.675	288.609	3463.302	285.569	3426.823
	R2	0.817	9.805	0.828	9.933	0.826	9.914	0.826	9.91
Bi-LSTM	RSME	335.629	4027.552	326.777	3921.326	326.845	3922.144	326.68	3920.155
	MAE	217.087	2605.041	201.043	2412.521	200.346	2404.157	198.599	2383.191
	R2	0.87	10.445	0.876	10.513	0.876	10.517	0.876	10.515
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**139.198**	**1670.371**	**136.844**	**1642.126**	**132.036**	**1584.437**	**137.381**	**1648.574**
	MAE	**89.05**	**1068.605**	**88.694**	**1064.327**	**84.339**	**1012.067**	**88.542**	**1062.503**
	R2	**0.971**	**11.655**	**0.97**	**11.64**	**0.972**	**11.669**	**0.972**	**11.663**
CEEMDAN-EWT-LSTM (Benchmark)	RSME	154.203	1850.436	151.214	1814.569	159.114	1909.362	162.702	1952.424
	MAE	104.191	1250.294	102.33	1227.956	107.294	1287.529	107.308	1287.691
	R2	0.959	11.507	0.955	11.463	0.95	11.394	0.955	11.464
CEEMDAN-EWT-Bi-LSTM	RSME	151.427	1817.125	150.957	1811.487	149.384	1792.609	179.811	2157.728
	MAE	99.676	1196.116	101.16	1213.921	99.631	1195.575	123.906	1486.87
	R2	0.965	11.581	0.956	11.468	0.958	11.5	0.933	11.19
CEEMDAN-EWT-Transformer	RSME	281.531	3378.368	279.807	3357.678	286.151	3433.811	369.661	4435.932
	MAE	205.216	2462.593	201.873	2422.47	211.082	2532.98	277.186	3326.231
	R2	0.909	10.905	0.911	10.934	0.907	10.88	0.827	9.921

Table 4.4: Forecasting Accuracy of Different Methods for Varying Input Lengths with a Fixed 30-Minute Horizon.

Model	Metrics	1 hr - 1 hr		1.5 hr - 1 hr		2 hr - 1 hr		4 hr - 1 hr	
		Average	Sum	Average	Sum	Average	Sum	Average	Sum
Transformer	RSME	474.08	5688.962	495.556	5946.667	390.979	4691.751	495.998	5951.979
	MAE	358.824	4305.892	378.976	4547.712	292.543	3510.517	382.669	4592.024
	R2	0.745	8.939	0.714	8.565	0.825	9.899	0.705	8.456
RNN	RSME	482.649	5791.788	469.162	5629.939	414.75	4976.997	798.335	9580.017
	MAE	357.561	4290.73	346.509	4158.11	311.356	3736.277	410.813	4929.759
	R2	0.735	8.82	0.741	8.893	0.804	9.648	0.648	7.774
LSTM	RSME	502.833	6033.991	484.157	5809.887	408.496	4901.947	481.456	5777.471
	MAE	383.028	4596.335	359.989	4319.869	304.631	3655.567	357.571	4290.854
	R2	0.714	8.568	0.738	8.855	0.81	9.716	0.739	8.862
GRU	RSME	463.591	5563.094	450.976	5411.717	386.333	4635.997	448.137	5377.647
	MAE	339.115	4069.385	330.991	3971.895	288.609	3463.302	325.449	3905.389
	R2	0.745	8.934	0.761	9.135	0.826	9.914	0.767	9.207
Bi-LSTM	RSME	407.983	4895.793	407.479	4889.746	326.845	3922.144	409.367	4912.4
	MAE	256.9	3082.802	255.636	3067.636	200.346	2404.157	254.975	3059.697
	R2	0.808	9.698	0.81	9.715	0.876	10.517	0.809	9.703
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**188.074**	**2256.891**	**182.877**	**2194.526**	**132.036**	**1584.437**	**181.742**	**2180.905**
	MAE	**121.369**	**1456.425**	**117.507**	**1410.087**	**84.339**	**1012.067**	**116.222**	**1394.665**
	R2	**0.951**	**11.408**	**0.952**	**11.424**	**0.972**	**11.669**	**0.958**	**11.494**
CEEMDAN-EWT-LSTM (Benchmark)	RSME	205.698	2468.378	216.75	2601.005	159.114	1909.362	217.527	2610.328
	MAE	140.196	1682.352	148.469	1781.633	107.294	1287.529	147.218	1766.614
	R2	0.934	11.204	0.912	10.942	0.95	11.394	0.919	11.024
CEEMDAN-EWT-Bi-LSTM	RSME	203.552	2442.623	207.312	2487.747	149.384	1792.609	222.171	2666.052
	MAE	137.051	1644.613	139.27	1671.239	99.631	1195.575	153.787	1845.441
	R2	0.937	11.239	0.929	11.144	0.958	11.5	0.924	11.088
CEEMDAN-EWT-Transformer	RSME	295.626	3547.513	336.074	4032.888	286.151	3433.811	416.765	5001.183
	MAE	212.008	2544.101	243.377	2920.522	211.082	2532.98	306.991	3683.897
	R2	0.902	10.823	0.874	10.492	0.907	10.88	0.799	9.582

Table 4.5: Forecasting Accuracy of Different Methods for Varying Input Lengths with a Fixed 1-Hour Horizon.

4.5.1 Experimental Setup

In this experiment, we evaluate the performance of the multi-feature model. The following steps outline the mathematical formulation of this model.

Step 1: Multi-Feature Input Definition

Define the multi-feature input at time t:

$$\mathbf{X}_t = \left[x_t^{(1)}, x_t^{(2)}, \ldots, x_t^{(N)}\right] \tag{4.7}$$

where:

- $\mathbf{X}_t$ represents the input vector at time t.
- $x_t^{(j)}$ denotes the value of the j-th feature at time t.
- N is the total number of features.

Step 2: Decomposing Each Feature Using CEEMDAN

Each feature $x_t^{(j)}$ is decomposed using CEEMDAN to obtain K IMFs:

$$x_t^{(j)} \xrightarrow{\text{CEEMDAN}} \{IMF_t^{(j,1)}, IMF_t^{(j,2)}, \ldots, IMF_t^{(j,K)}\} \tag{4.8}$$

where $IMF_t^{(j,k)}$ represents the k-th IMF of feature j at time t.

Step 3: Processing IMF1 Using EWT

For each feature j, EWT to the first IMF $IMF_t^{(j,1)}$:

$$IMF_t^{(j,1)} \xrightarrow{\text{EWT}} I\tilde{M}F_t^{(j,1)} \tag{4.9}$$

where $I\tilde{M}F_t^{(j,1)}$ is the transformed version of $IMF_t^{(j,1)}$ after EWT processing.

Step 4: Stacking IMFs of the Same Index Across Features

For each IMF index k ($k \in \{1, 2, \ldots, K\}$), construct a combined input by stacking IMFs across all features:

$$\mathbf{IMF}_t^{(k)} = \left[IMF_t^{(1,k)}, IMF_t^{(2,k)}, \ldots, IMF_t^{(N,k)}\right] \tag{4.10}$$

For $k = 1$, replace $IMF_t^{(j,1)}$ with its EWT-transformed version $I\tilde{M}F_t^{(j,1)}$:

$$\mathbf{IMF}_t^{(1)} = \left[I\tilde{M}F_t^{(1,1)}, I\tilde{M}F_t^{(2,1)}, \ldots, I\tilde{M}F_t^{(N,1)}\right] \tag{4.11}$$

Step 5: Training GRU for Each IMF Independently

Each stacked IMF group $\mathbf{IMF}_t^{(k)}$ is used as input to a separate GRU model:

$$\hat{y}_t = \sum_{k=1}^{K} f_{\text{GRU}}^{(k)}(\mathbf{IMF}_t^{(k)}) \tag{4.12}$$

where $f_{\text{GRU}}^{(k)}(\cdot)$ is the GRU model corresponding to the IMF index k.

In Table 4.6, **CEEMDAN-EWT-GRU (Proposed Method - All Feature)** refers to adding all the features in the proposed method. The **CEEMDAN-EWT-GRU (Proposed Method - 3 Feature)** refers to adding the wind power, wind speed and theoretical power curve. **CEEMDAN-EWT-GRU (Proposed Method - 2 Feature)** and **CEEMDAN-EWT-GRU-MHA (2 Feature)** refer to the features of both the wind power and the theoretical power curve.

4.5.2 Results and Discussion

Based on the results from 4.6 for comparing the single-feature proposed method vs. the multi-feature proposed methods, it was assessed using RMSE, MAE, and R^2 scores across twelve months to determine forecasting accuracy. The proposed method based on a single feature performs best by obtaining the lowest RMSE and MAE values, with 92.053 and 59.831, respectively. Also, it maintains a high value of the R^2 score (0.989), indicating strong predictive capability. On the other hand, the proposed method with all features did not achieve the expectation by achieving the worst performance, getting the highest RMSE (956.15) and a negative R^2 (-0.114). The same result happened to the method with three features by getting RMSE (971.19) and a negative R^2 (-0.11), indicating that reducing the number of features only slightly mitigated the issue. From the results, adding the features impacts the performance of the proposed negatively; this is related to the nature of the decomposition technique of CEEMDAN since CEEMDAN can decompose a univariate/ single feature only without considering their relationship with wind power. As a result, the IMF from different features does not align in frequency, which leads to mixed information that makes it difficult for the GRU model to learn functional patterns as the mixed information of the frequency has added more noise, further reducing the accuracy. Also, adding the Multi-Head Attention did not solve the issue; it got worse by resulting in an RMSE of 1046.65, MAE of 878.55, and a negative R^2 (-0.276), indicating that the additional complexity did not contribute to better forecasting since the issue is the mixed information of IMFs. This aligns with the findings of the paper [33], which highlighted that independent decomposition of multi-time series often leads to suboptimal feature extraction due to a lack of shared frequency components. Interestingly, the two-feature variant of the CEEMDAN-EWT-GRU model demonstrated a strong balance between accuracy and model stability. With an RMSE of 140.41, MAE of 85.73, and a solid R^2 score of 0.972, the two features were similar, Wind power and Theortrical Power Curve, which both contain power (KW), but there is a difference between them in computing the power, so the mixing of the frequency of the feature are not different that result as stable and accuracy but not outperform the proposed method CEEMDAN-EWT-GRU of only one single feature. In

Model	Metrics	Jan	Feb	March	April	May	June	July	August	Sept	Oct	Nov	Dec	Average	Sum
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**159.848**	**104.645**	**115.854**	**65.954**	**120.018**	**129.677**	**44.944**	**80.535**	**96.789**	**64.776**	**85.281**	**36.311**	**92.053**	**1104.632**
	MAE	**113.258**	**62.112**	**67.037**	**41.752**	**81.743**	**94.476**	**25.51**	**54.866**	**70.223**	**45.104**	**40.109**	**21.787**	**59.831**	**717.977**
	R2	**0.95**	**0.993**	**0.993**	**0.997**	**0.983**	**0.983**	**0.992**	**0.995**	**0.994**	**0.997**	**0.992**	**0.994**	**0.989**	**11.863**
CEEMDAN-EWT-GRU (Proposed Method - All Feature)	RSME	220.351	414.72	1491.743	679.698	1131.281	1176.162	211.185	1259.943	1413.096	1107.175	1988.602	379.871	956.152	11473.83
	MAE	104.315	282.335	1188.315	520.971	1069.743	1073.969	129.809	1019.686	1311.468	931.744	1918.187	141.817	807.697	9692.359
	R2	0.904	0.884	-0.08	0.681	-0.496	-0.439	0.822	-0.272	-0.257	-0.238	-3.176	0.297	-0.114	-1.37
CEEMDAN-EWT-GRU (Proposed Method - 3 Feature)	RSME	285.518	390.225	1485.191	695.611	1129.929	1198.496	201.165	1233.679	1452.335	1156.259	2048.36	377.559	971.194	11654.33
	MAE	174.101	266.761	1183.502	530.815	1068.44	1092.821	125.434	1002.032	1346.041	968.972	1976.164	141.306	823.032	9876.389
	R2	0.839	0.898	-0.07	0.666	-0.492	-0.495	0.839	-0.22	-0.328	0.169	-3.431	0.306	-0.11	-1.319
CEEMDAN-EWT-GRU (Proposed Method - 2 Feature)	RSME	212.009	163.802	186.19	92.857	191.646	180.967	71.852	100.883	153.447	126.395	120.295	84.537	140.407	1684.88
	MAE	82.71	104.13	111.074	58.194	127.486	127.681	48.117	66.742	108.805	85.062	55.376	53.457	85.736	1028.834
	R2	0.911	0.982	0.983	0.994	0.957	0.966	0.979	0.992	0.985	0.99	0.965	0.965	0.972	11.669
CEEMDAN-EWT-GRU-MHA (2 Feature)	RSME	271.12	414.595	1506.232	792.722	1174.452	1260.331	229.526	1302.368	2052.557	1090.491	2069.477	395.905	1046.648	12559.78
	MAE	179.773	281.311	1197.858	588.59	1105.397	1143.351	135.803	1047.421	1803.873	917.705	1996.327	145.228	878.553	10542.64
	R2	0.855	0.884	-0.101	0.566	-0.612	-0.653	0.79	-0.359	-1.653	0.261	-3.523	0.237	-0.276	-3.308

Table 4.6: Accuracy of the Proposed Method with Varying Feature Counts for a 1-Hour Historical Length and 10-Minute Forecasting Horizon.

conclusion, the CEEMDAN-EWT-GRU baseline provides its performance compared to another method with excessive features that result in worse performance due to the nature of the CEEMDAN, which is not equivalent to multi-feature models.

4.6 Experiment 3: Hybrid GRU-BiLSTM Model Based on Frequency Decomposition

This experiment investigates a framework inspired by [37], in which the IMFs have been split into high-frequency and low-frequency components using permutation entropy and feeding them into separate deep learning models if it improves forecast accuracy. The evaluation of the hybrid method will cover all the time horizons covered in the first experiment.

4.6.1 Experimental Setup

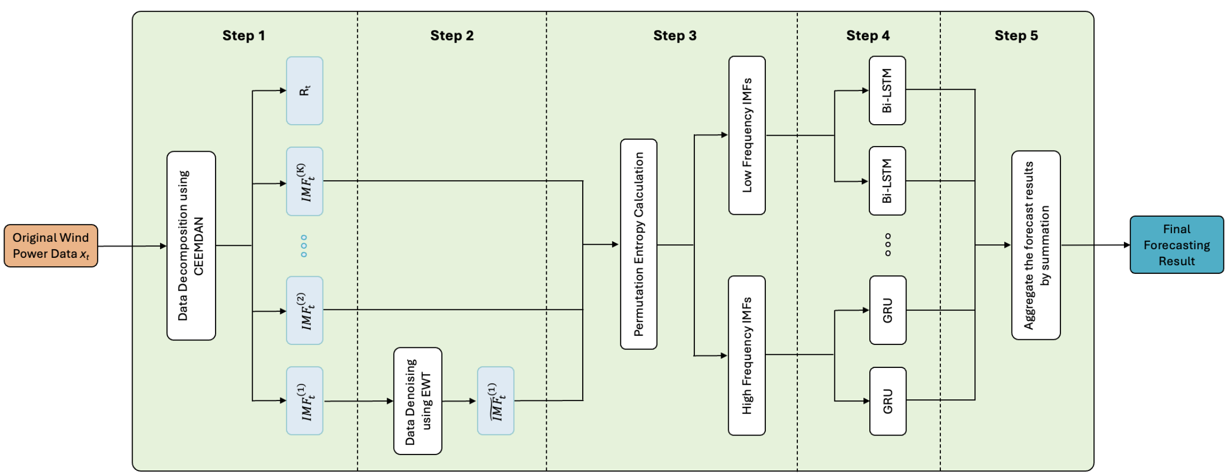

Figure 4.2: Framework of the CEEMDAN-EWT-GRU-BiLSTM Model.

Figure 4.2 is the framework for CEEMDAN-EWT-GRU-BiLSTM. First, the original data will go through the CEEMDAN. After obtaining the IMFs, apply the EWT on the first IMF. The IMF outputs from the previous step are then used to calculate the permutation entropy (PE) values [1]. Based on these values, IMFs are divided into high-frequency and low-frequency IMFs by a fixed threshold. IMFs classified as high-frequency will be fed to the GRU since it can handle the high-frequency components, and the rest of the IMFs considered low-frequency components are fed into the Bi-LSTM.

Permutation entropy (PE) is a complexity measure for time series introduced by Bandt et al.[1]. It quantifies the diversity of ordinal patterns in the time series.

The **Permutation Entropy** (PE) for an Intrinsic Mode Function (IMF) is given by [1]:

$$H_i(d) = -\sum_{j=1}^{N_d} p_j \log p_j \tag{4.13}$$

where:

- d is the embedding dimension.
- $N_d = d!$ is the total number of possible ordinal patterns.
- p_j is the probability of the j-th ordinal pattern.

To normalize the entropy, use:

$$H_n(d) = \frac{H_i(d)}{\log N_d} \tag{4.14}$$

where $H_n(d)$ represents the normalized permutation entropy.

To classify the IMF_i as either **high-frequency** or **low-frequency**, we define a threshold $T = 0.45$ and apply the following rule:

$$\mathrm{IMF}_i = \begin{cases} \text{High-Frequency}, & \text{if } H_n(d) > T \\ \text{Low-Frequency}, & \text{if } H_n(d) \leq T \end{cases} \tag{4.15}$$

4.6.2 Results and Discussion

The results in Table 4.7 show that the proposed method outperforms other approaches, including CEEMDAN-EWT-GRU-BiLSTM, on multiple evaluation metrics such as RMSE, MAE, R^2, and Margin of Error. This improvement is observed for an input length of 1 hour and a forecasting horizon of 10 minutes, highlighting the optimal input and forecasting configuration. Furthermore, The results in Table 4.8 compare the proposed CEEMDAN-EWT-GRU method with the hybrid CEEMDAN-EWT-GRU-BiLSTM approach across varying forecasting horizons. Evaluation metrics, including the RMSE, MAE, and R^2 score, were analyzed for different time windows. For shorter forecasting horizons, in particular, 10 min, as the historical length is from 1 hour to 4 hours, the proposed method CEEMDAN-EWT-GRU has shown surpasses the hybrid method by achieving lower RMSE and MAE values, which indicates better prediction accuracy. As we increase the forecasting horizon, such as 20 minutes, 30 minutes, and 1 hour, both models increase in error for the RMSE and MAE. However, the proposed method maintained lower values in RMSE, and MAE than the hybrid models, suggesting better generalizability for wind power forecasting for any time horizon.

Introducing the Bi-LSTM to extract patterns from low-frequency IMFs did not exceed the expectation of better performance compared to the proposed method that utilizes only

Model	Metrics	Jan	Feb	March	April	May	June	July	August	Sept	Oct	Nov	Dec	Average	Sum
CEEMDAN-EWT-GRU (Proposed Method)	RSME	159.848	**104.645**	**115.854**	**65.954**	**120.018**	**129.677**	**44.944**	**80.535**	**96.789**	**64.776**	85.281	36.311	92.053	1104.632
	MAE	113.258	**62.112**	**67.037**	**41.752**	**81.743**	**94.476**	**25.51**	**54.866**	**70.223**	**45.104**	40.109	21.787	59.831	717.977
	R2	0.95	**0.993**	**0.993**	**0.997**	**0.983**	**0.983**	**0.992**	**0.995**	**0.994**	**0.997**	0.992	0.994	0.989	11.863
	Margin of Error	9.559	**7.226**	**7.592**	**4.356**	**7.874**	**8.754**	**2.924**	**5.314**	**6.298**	**4.367**	6.077	2.293	6.053	72.634
CEEMDAN-EWT-Bi-LSTM	RSME	185.342	115.757	125.428	67.587	125.268	133.228	46.435	84.128	143.651	74.643	86.547	107.076	107.924	1295.09
	MAE	127.545	73.032	76.937	44.338	84.239	97.899	28.391	61.664	114.453	55.023	41.815	81.333	73.889	886.669
	R2	0.932	0.991	0.992	0.997	0.982	0.982	0.991	0.994	0.987	0.997	0.992	0.944	0.982	11.781
	Margin of Error	11.196	8.008	8.238	4.464	8.219	8.972	2.942	5.369	7.424	5.071	6.169	5.329	6.783	81.401
CEEMDAN-EWT-GRU-Bi-LSTM	RSME	**128.312**	112.318	126.333	67.041	120.39	130.45	45.165	81.961	144.624	75.907	86.881	71.282	99.222	1190.664
	MAE	**73.786**	70.774	78.193	43.143	81.953	95.362	25.85	58.272	114.716	55.226	43.048	53.429	66.146	793.752
	R2	**0.967**	0.992	0.992	0.997	0.983	0.982	0.992	0.995	0.987	0.996	0.992	0.975	0.988	11.85
	Margin of Error	8.762	7.743	8.314	4.468	7.921	8.805	2.93	5.34	7.506	4.808	6.2	3.701	6.375	76.498
CEEMDAN-EWT-GRU (Benchmark)	RSME	202.458	108.26	121.099	67.965	124.621	132.089	45.811	81.868	151.252	72.272	**84.338**	36.726	102.397	1228.759
	MAE	146.862	68.415	70.416	42.866	84.172	97.61	26.578	56.907	120.283	52.647	**42.483**	22.549	69.316	831.788
	R2	0.919	0.992	0.993	0.997	0.982	0.982	0.992	0.995	0.986	0.997	**0.992**	0.993	0.985	11.82
	Margin of Error	11.404	7.414	7.937	4.549	8.155	8.915	2.959	5.363	7.775	4.859	**6.015**	2.355	6.475	77.7

Table 4.7: Accuracy of the Hybrid Method for a 1-Hour Historical Length and 10-Minute Forecasting Horizon.

Model	Metrics	1 hr - 10 min		1.5 hr - 10 min		2 hr - 10 min		4 hr - 10 min	
		Average	Sum	Average	Sum	Average	Sum	Average	Sum
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**92.053**	**1104.632**	**95.485**	**1145.823**	**94.319**	**1131.832**	**93.716**	**1124.59**
	MAE	**59.831**	**717.977**	**63.54**	**762.48**	**63.108**	**757.295**	**61.491**	**737.893**
	R2	**0.989**	**11.863**	0.985	11.821	**0.987**	**11.839**	**0.987**	**11.838**
CEEMDAN-EWT-GRU-Bi-LSTM	RSME	99.222	1190.664	98.532	1182.378	106.729	1280.746	106.049	1272.591
	MAE	66.146	793.752	66.674	800.091	73.404	880.853	73.297	879.565
	R2	0.988	11.85	**0.988**	**11.852**	0.976	11.707	0.982	11.789
		1 hr - 20 min		1.5 hr - 20 min		2 hr - 20 min		4 hr - 20 min	
		Average	Sum	Average	Sum	Average	Sum	Average	Sum
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**124.536**	**1494.432**	**118.795**	**1425.542**	**114.656**	**1375.873**	**114.426**	**1373.115**
	MAE	**80.137**	**961.648**	**76.893**	**922.714**	**71.414**	**856.967**	**72.315**	**867.777**
	R2	**0.975**	**11.696**	**0.978**	**11.738**	**0.98**	**11.764**	**0.981**	**11.772**
CEEMDAN-EWT-GRU-Bi-LSTM	RSME	138.915	1666.982	126.096	1513.157	138.688	1664.256	138.124	1657.489
	MAE	93.401	1120.81	84.871	1018.449	96.105	1153.265	93.199	1118.386
	R2	0.967	11.604	0.973	11.679	0.96	11.521	0.965	11.584
		1 hr - 30 min		1.5 hr - 30 min		2 hr - 30 min		4 hr - 30 min	
		Average	Sum	Average	Sum	Average	Sum	Average	Sum
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**139.198**	**1670.371**	**136.844**	**1642.126**	**132.036**	**1584.437**	**137.381**	**1648.574**
	MAE	**89.05**	**1068.605**	**88.694**	**1064.327**	**84.339**	**1012.067**	**88.542**	**1062.503**
	R2	**0.971**	**11.655**	**0.97**	**11.64**	**0.972**	**11.669**	**0.972**	**11.663**
CEEMDAN-EWT-GRU-Bi-LSTM	RSME	161.839	1942.069	146.122	1753.464	148.782	1785.378	162.457	1949.484
	MAE	111.592	1339.103	97.811	1173.726	99.895	1198.738	110.318	1323.821
	R2	0.955	11.458	0.96	11.523	0.957	11.487	0.952	11.424
		1 hr - 1 hr		1.5 hr - 1 hr		2 hr - 1 hr		4 hr - 1 hr	
		Average	Sum	Average	Sum	Average	Sum	Average	Sum
CEEMDAN-EWT-GRU (Proposed Method)	RSME	**188.074**	**2256.891**	**182.877**	**2194.526**	**132.036**	**1584.437**	**181.742**	**2180.905**
	MAE	**121.369**	**1456.425**	**117.507**	**1410.087**	**84.339**	**1012.067**	**116.222**	**1394.665**
	R2	**0.951**	**11.408**	**0.952**	**11.424**	**0.972**	**11.669**	**0.958**	**11.494**
CEEMDAN-EWT-GRU-Bi-LSTM	RSME	197.182	2366.188	192.179	2306.142	196.204	2354.444	212.666	2551.994
	MAE	132.114	1585.368	127.568	1530.821	131.663	1579.961	142.97	1715.64
	R2	0.942	11.302	0.942	11.304	0.935	11.217	0.934	11.208

Table 4.8: Comparison of Forecasting Accuracy Between the Hybrid Method and Proposed Method Across All Time Horizons.

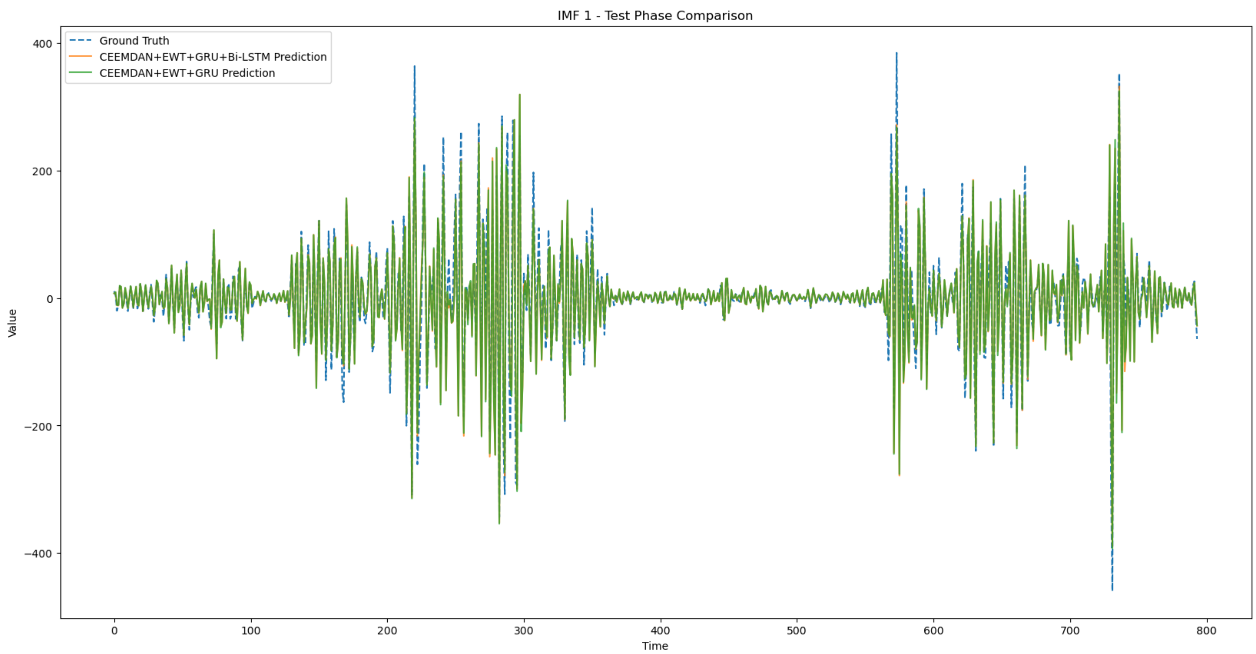

Figure 4.3: IMF1: High-Frequency Component.

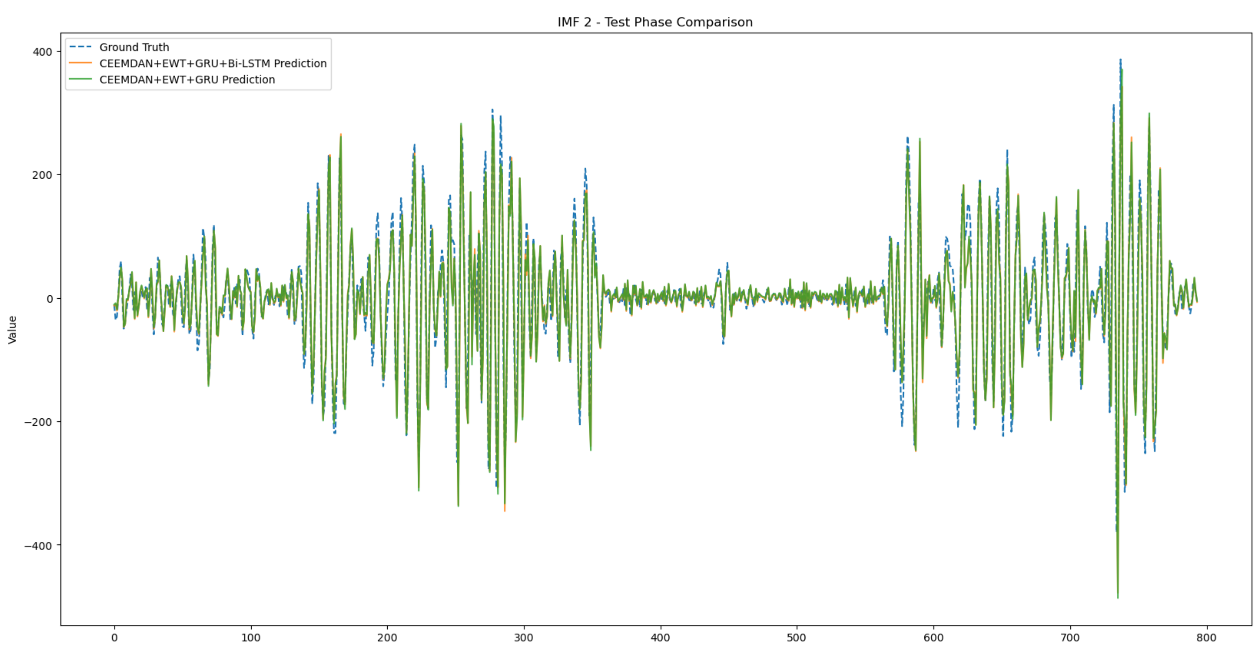

Figure 4.4: IMF2: High-Frequency Component.

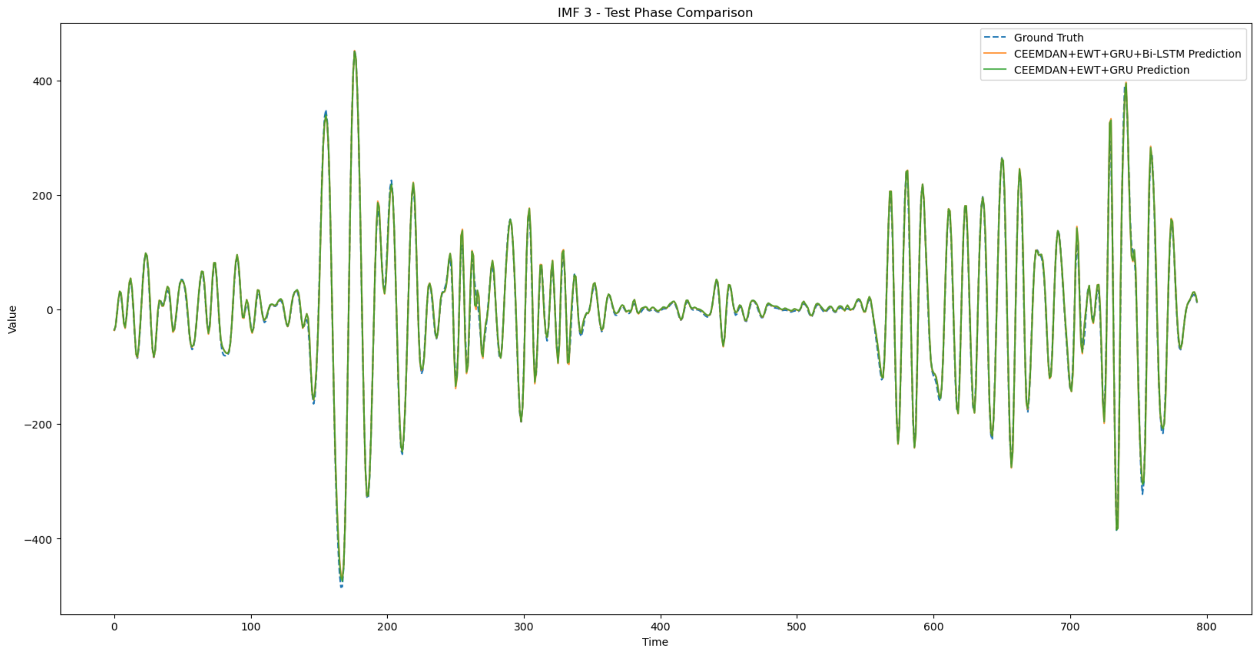

Figure 4.5: IMF3: High-Frequency Component.

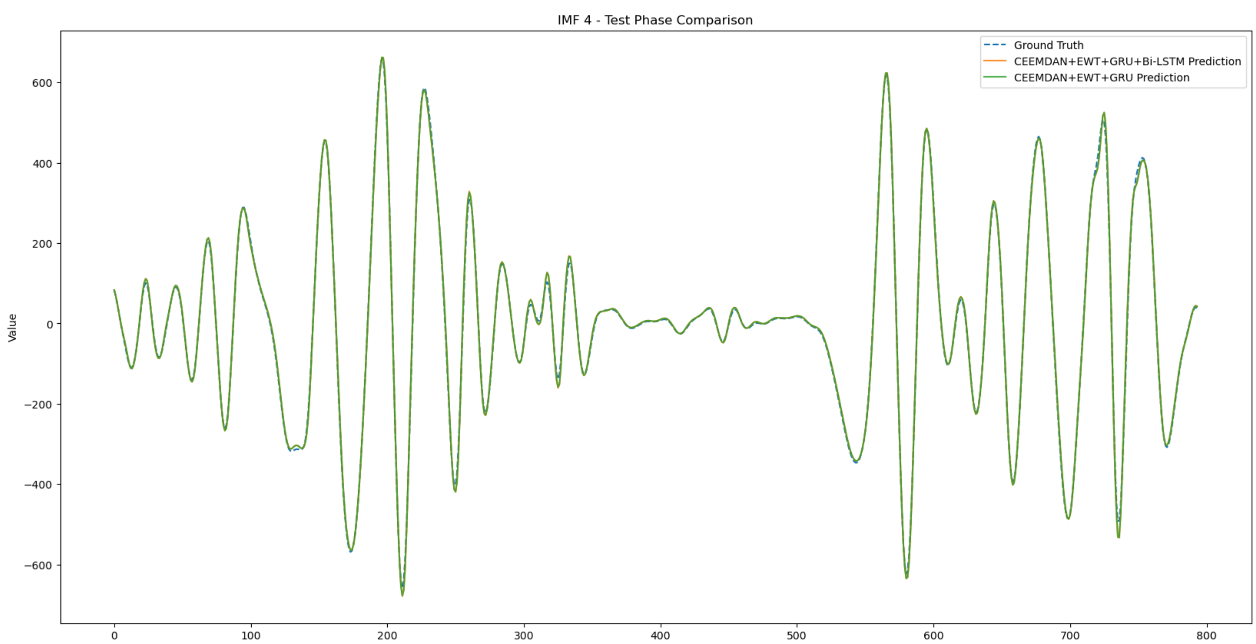

Figure 4.6: IMF4: High-Frequency Component.

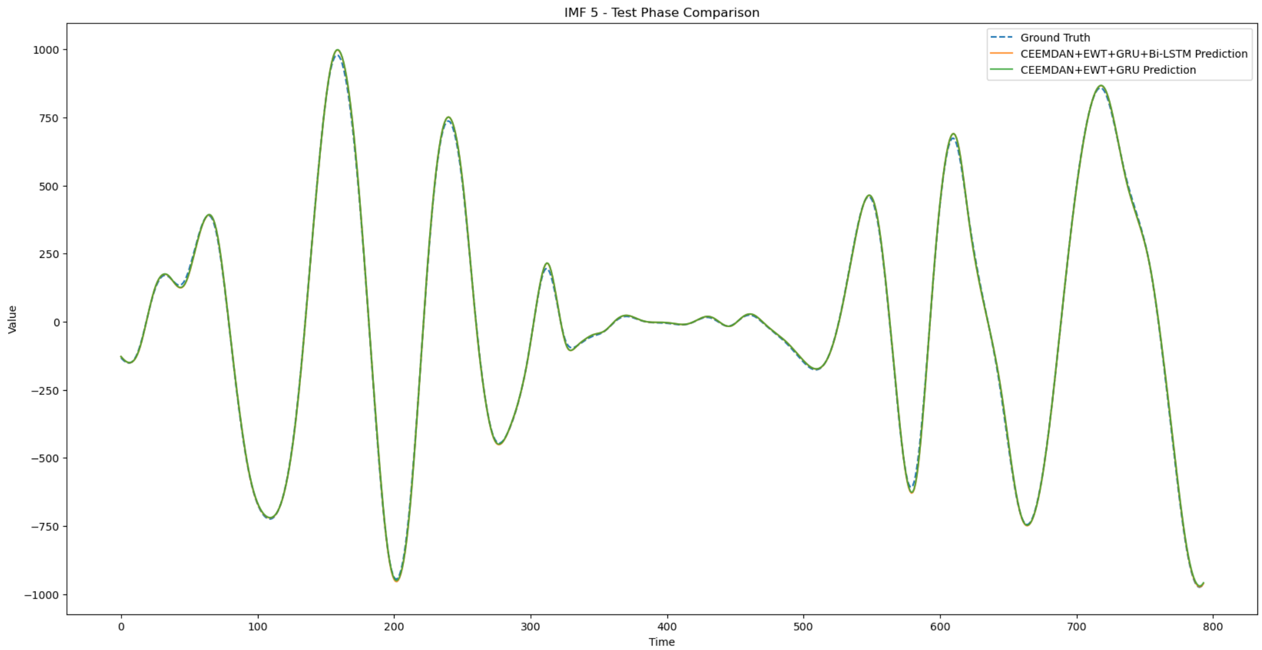

Figure 4.7: IMF5: High-Frequency Component.

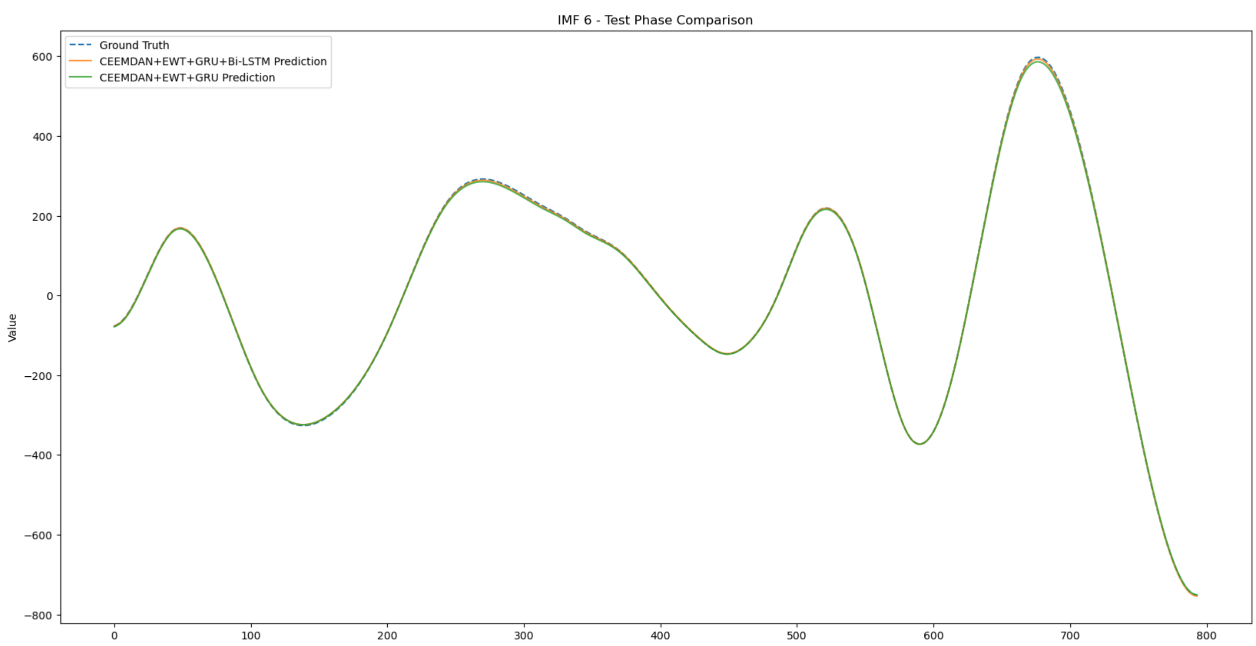

Figure 4.8: IMF6: Low-Frequency Component.

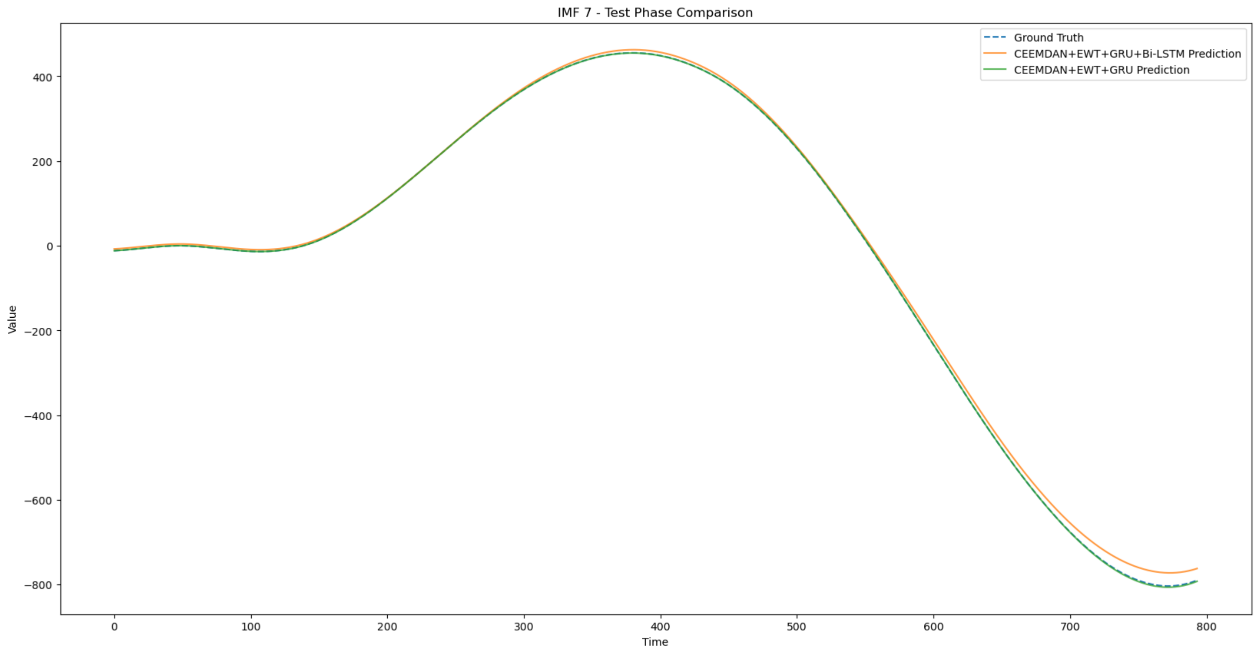

Figure 4.9: IMF7: Low-Frequency Component.

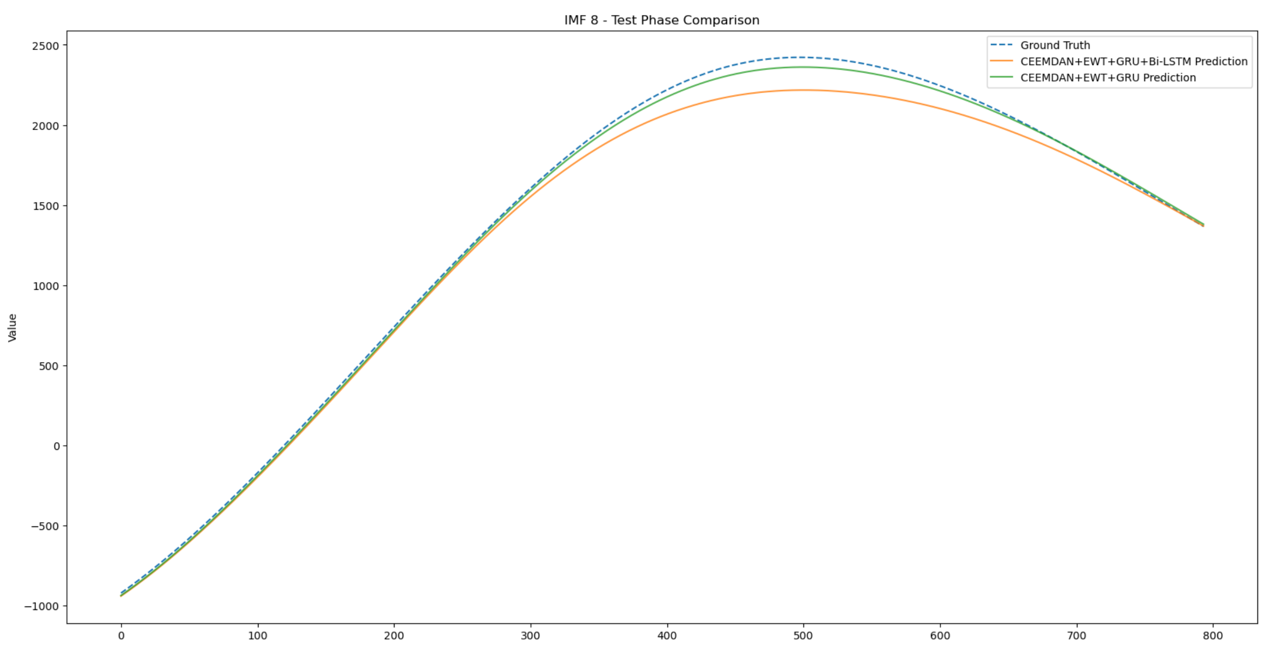

Figure 4.10: IMF8: Low-Frequency Component.

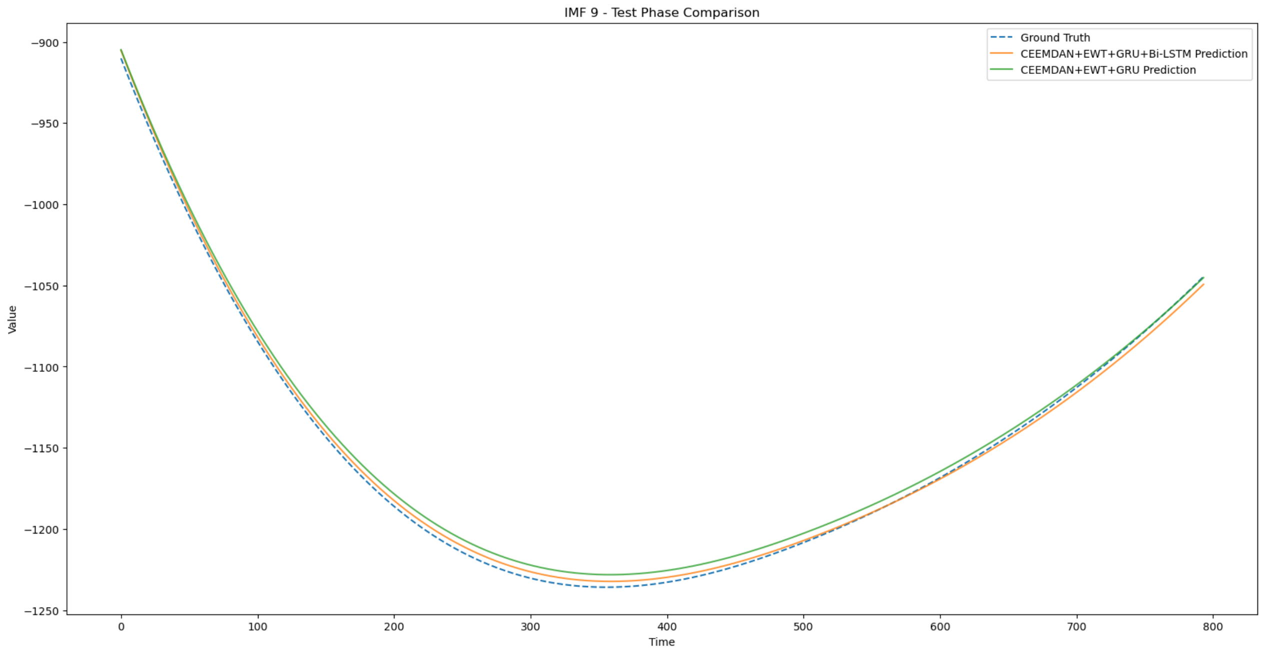

Figure 4.11: IMF9: Low-Frequency Component.

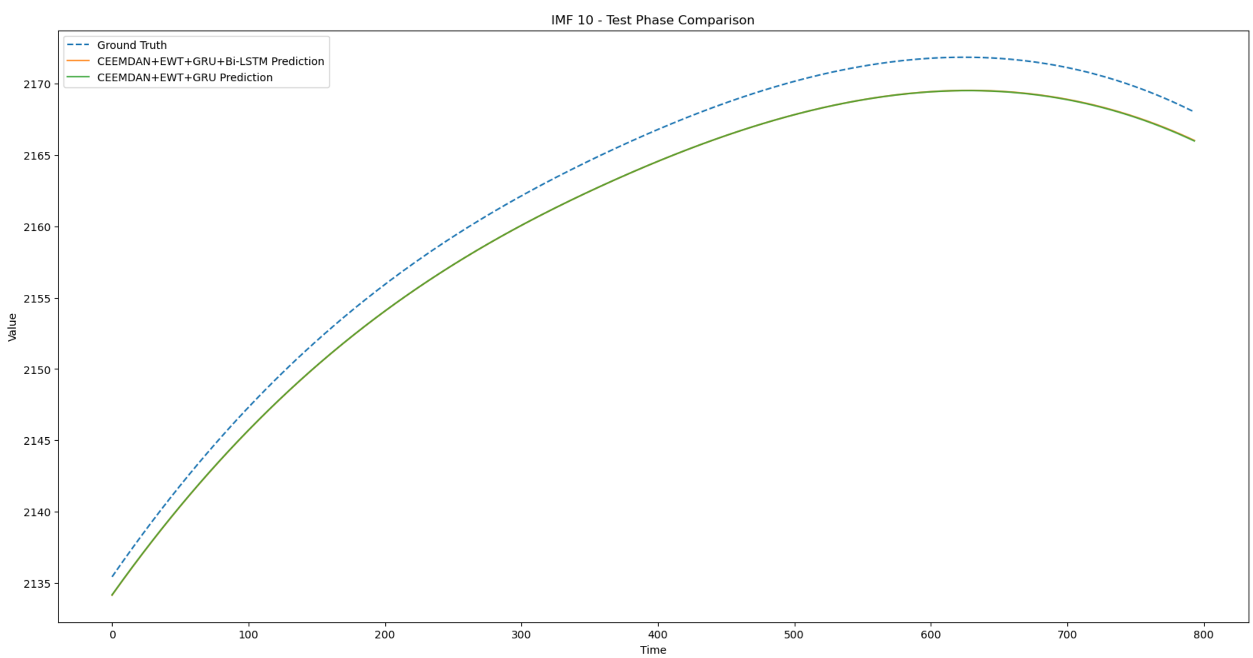

Figure 4.12: IMF10: Low-Frequency Component.

one deep learning model. Next, we will investigate on the basis of the prediction of the proposed method and hybrid methods for each IMF to see how Bi-LSTM reduced the prediction accuracy as we introduced it in the framework.

Both methods have performed well in the high-frequency IMFs from Figure 4.3 to Figure 4.7 compared to the ground truth, suggesting how effective the GRU is at handling high-frequency IMFs.Furthermore, when looking for low-frequency IMFs from Figure 4.8 to Figure 4.12, we notice that there is a gap between the proposed method CEEMDAN-EWT-GRU and the hybrid method CEEMDAN-EWT-GRU-BiLSTM. The IMF 7-10 GRU in the proposed method effectively captures the trend, but there is a struggle for the Bi-LSTM in the hybrid method, which shows it fails to capture the trend of low-frequency IMFs. Hence, integrating Bi-LSTM in the method did not enhance the prediction accuracy, but the opposite was achieved. The reason Bi-LSTM's bidirectional structure has introduced complexity is that it demands more parameters and memory requirements since the bidirectional structure makes it prone to redundant information and overemphasizing the past information, which is the opposite of the nature of GRU that a simpler gate, fewer parameters, and state updates allow it to effectively model the high-frequency and low-frequency IMFs, so making it a perfect combination with the CEEMDAN+EWT.

In general, from the results 4.8, the proposed method is more robust and reliable wind power forecasting due to the ability of the GRU to capture patterns from both high-frequency and low-frequency. The addition of the Bi-LSTM did not enhance the performance of the prediction.

4.7 Discussion and Analysis

4.7.1 Insights from Results

This research study conducted three experiments, each with unique objectives. Overall, the proposed method has demonstrated that it has effectiveness in all experiments against the benchmark [17] and hybrid methods. The first experiment evaluates the proposed method for specific time horizons detailed in Experiment 1 and compares its performance against other forecasting methods. The proposed method has achieved the best performance compared to others, since it uses the GRU that provides benefits compared to the benchmark [17], in which the GRU has a simple architecture, fewer parameters and training time and can effectively handle high-frequency components.

The second experiment aims to integrate more features into the proposed method to evaluate their performance against the proposed method with a single feature. The results have indicated that incorporating more features in the proposed method has resulted in the opposite of the expectation because it performs worse than the proposed method with a single feature. The reason is that the CEEMDAN is designed to decompose only feature individuals, which causes a different frequency between the features for the same IMF index, which leads to mixing information of frequency, which causes the GRU to have difficulty understanding the pattern.

The third experiment introduces a hybrid method composite of CEEMDAN-EWT-GRU-BiLSTM inspired from [37], in which the high-frequency and low-frequency IMFs from CEEMDAN are processed separately using GRU and Bi-LSTM. This new hybrid method did not outperform the proposed method due to the Bi-LSTM, which did not effectively capture the pattern in low-frequency IMFs because it introduces more complexity, parameters, and memory, which leads to the preservation of redundant information.

Overall, the proposed method CEEMDAN-EWT-GRU has demonstrated effectiveness and is significantly performed in different experiments with different experimental setups. The reasons for its effectiveness have been stated in the previous section.

4.7.2 Strengths and Limitations

This section focuses on the proposed framework's strengths and limitations. Based on the experiment results, the proposed framework has three strengths:

- **Noise Reduction and Feature Extraction:** Utilizing both CEEMDAN and EWT has shown benefits, such as extracting meaningful features while reducing noise, which enhances prediction accuracy. These techniques effectively handle the issue of non-stationary wind data.
- **Improved Forecasting Accuracy:** The proposed method has shown superiority over other forecasting methods, which demonstrates its effectiveness in dealing with the key challenge of wind data.
- **Scalability and Adaptability:** The proposed method outperforms other methods for different input and output configurations, making it suitable for different time horizons.

The proposed method is only capable of a single feature; this is considered a limitation of the proposed framework. The CEEMDAN is built to decompose a single feature; this limits to multivariate time series forecasting scenarios.

4.7.3 Future Research Directions

Recalling from the second experiment the proposed framework well perform for single feature. The reason that CEEMDAN is designed to decompose a single feature is that we have decomposed each feature independently, which leads to a different frequency of the same IMF indices. That means a misalignment in the frequency of the same IMF index, creating higher noise and complexity so that the model will not learn this noise, causing poor performance. This is why to tackle the issue of the second experiment by utilizing multivariate decomposition methods. Also, future work should investigate performance improvements over longer horizons. The reason is that we solve the issue of performance drop in prediction for longer horizons. This can lead to optimizing energy management since we can predict for long periods. The last thing to focus on is to evaluate the proposed method with many wind forecasting datasets, trying to observe the performance of the proposed method in different wind datasets can give us how is robustness to handle different wind datasets that might be in different locations and periods.

Chapter 5

Conclusion

The ever-increasing demand for nonrenewable energy has negatively impacted our environment. To counteract this, the government's vision has been to utilize a source of energy that provides sustainability and clean energy for our environment. Wind energy is a renewable energy that provides clean energy and sustainability compared to other nonrenewable energy. However, the inherent characteristics of wind make it difficult to predict wind power. This thesis has reviewed several wind power forecasting methods, such as physical modeling, statistical methods, machine learning, and hybrid methods integrated with decomposition techniques. NWP model is a physical model that utilizes atmospheric and physical laws for forecasting. However, the weakness of this method is that it does not consider historical trends or capture localized wind behavior. So, the introduction of statistics has tackled the limitations of physical modeling by considering historical trends, such as methods such as AR, ARMA, sVAR, etc. However, the most significant drawback of this method is that they are not suited for non-linear relationships.

ML has solved the limitations of conventional methods through its ability to analyze and forecast complex patterns within data, enabling the identification of intricate relationships and adaptation to non-linear relationships. Traditional ML and DL show more effectiveness in wind power forecasting than conventional methods. However, the key limitation of these methods is that they struggle to deal with non-stationary time series, characteristic of wind effectively. Research papers have focused on using decomposition techniques such as WD and MD with deep learning models. Those techniques are effective when dealing with non-stationary time series, which breaks down the original time series into multiple relatively stable subsequences. The integration of methods has shown effectiveness in addressing these issues. However, as the wind often contains significant noise, advancing decomposition should be necessary to improve wind power forecasting. Advanced decomposition, such as CEEMDAN and EWT, offers a promising focus for dealing with nonstationary and noisy wind power data. Hence, we have proposed a method that contains an advanced method with a deep learning model, such as CEEMDAN-EWT-GRU, to enhance prediction accuracy. There were three experiments to analyze the performance of the proposed method with different objectives and experimental setups.

The main objective of the first experiment is to compare the proposed method against baselines, benchmark (CEEMDAN-EWT-LSTM) and another hybrid method such as the CEEMDAN-EWT-Transformer. This experiment setup consists of 16 input and output configurations. The findings of the experiment show that the proposed method shows superiority over other methods for varying time horizons, showcasing the beneficial combination of CEEMDAN and EWT with the GRU model. The reason is that it is a simpler architecture with fewer gates and parameters that cause the GRU to focus more on the temporal dependencies of the decomposed data. In addition, the GRU is effective in dealing with high-frequency IMFs.

The second experiment investigated the impact of incorporating additional features, such as wind speed and direction, in the proposed method to observe its performance. Adding more features had the opposite effect due to information mixing across the frequency of the IMF index, as CEEMDAN can decompose each feature individually. The results of the second experiment suggest that the proposed method performs effectively with a single feature.

The third experiment introduces a new architecture similar to the proposed method. However, the difference is that we split the IMFs into high- and low-frequency components using permutation entropy, in which the GRU will be trained on the high-frequency components, and Bi-LSTM will be trained on low-frequency components and then compared to the proposed methods. However, the new hybrid method CEEMDAN-EWT-GRU-BiLSTM did not outperform the proposed method across different time horizons, primarily because of the introduction of the Bi-LSTM. The ability of Bi-LSTM to manage more training parameters and memory requirements allowed it to retain more redundant information compared to GRU, which is considered simple and efficient with CEEMDAN and EWT.

The proposed method consistently outperformed others in nearly all cases and metrics in all three experiments, suggesting that it could significantly improve wind power predictions. However, the proposed method has a limitation: It is effective for only one feature, since CEEMDAN focuses primarily on the decomposition of individual signal components. Hence, future works should investigate decomposition techniques suited for multivariate time series cases. Furthermore, future research should evaluate the proposed framework for different wind power datasets.

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APPENDICES

Appendix A

Proof of Concept

Appendix B

Python Implementation

B.1 Libraries

B.2 Code